

HITACHI

SERVICE MANUAL

TECHNICAL INFORMATION

SM0016

RAS-07BH1/RAC-07BH1
RAS-09BH1/RAC-09BH1
RAS-14BH1/RAC-14BH1

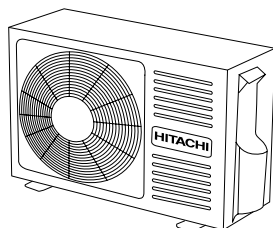
REFER TO THE FOUNDATION MANUAL

FOR SERVICE PERSONNEL ONLY

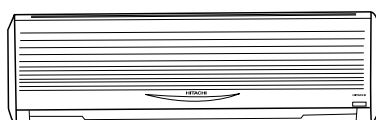
OUTDOOR UNIT

INDOOR UNIT

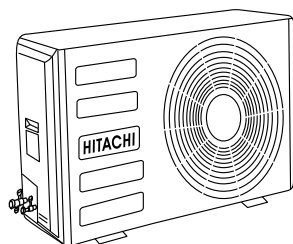
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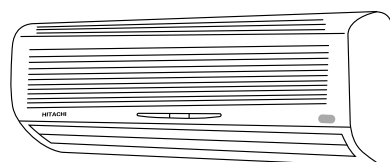
RAC-07BH1
RAC-09BH1



RAS-07BH1
RAS-09BH1



RAC-14BH1



RAS-14BH1

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SPECIFICATIONS

TYPE			WALL TYPE				
			COOLING UNIT		CONDENSING UNIT	COOLING UNIT	CONDENSING UNIT
MODEL			RAS-07BH1 RAS-09BH1		RAC-07BH1 RAC-09BH1	RAS-14BH1	RAC-14BH1
POWER SOURCE			1ϕ, 220 V, 50 Hz				
COOLING	TOTAL INPUT	(W)	850 - 875			1,130	
	TOTAL AMPERES	(A)	3.95 - 4.06			5.25	
	CAPACITY	(kW)	2.25 - 2.60			3.60	
		(BTU/H)	7,679 - 8,874			12,287	
HEATING	TOTAL INPUT	(W)	820 - 840			1300	
	TOTAL AMPERES	(A)	3.81 - 3.90			6.03	
	CAPACITY	(KW)	2.50 - 2.90			4.20	
		(BTU/H)	8,533 - 9,898			14,335	
DIMENSIONS (mm)		W	785	775	815	820	
		H	250	500	298	520	
		D	175	300	194	280	
NET WEIGHT (kg)			9	30.5	9	38	

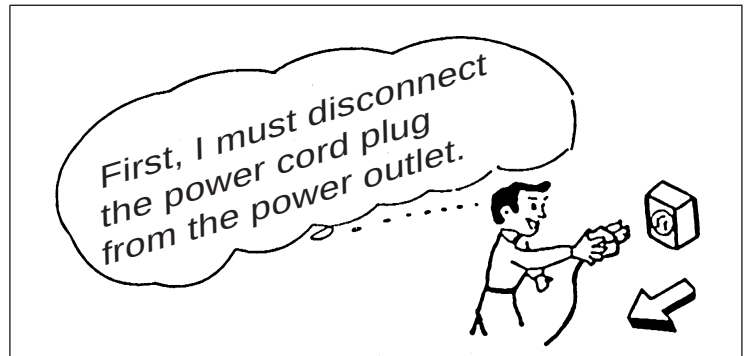
SPECIFICATIONS AND PARTS ARE SUBJECT TO CHANGE FOR IMPROVEMENT

ROOM AIR CONDITIONER

COOLING UNIT + CONDENSING UNIT

SAFETY DURING REPAIR WORK

1. In order to disassemble and repair the unit in question, be sure to disconnect the power cord plug from the power outlet before starting the work.



2. If it is necessary to replace any parts, they should be replaced with respective genuine parts for the unit, and the replacement must be effected in correct manner according to the instructions in the Service Manual of the unit.

If the contacts of electrical parts are defective, replace the electrical parts without trying to repair them.



3. After completion of repairs, the initial state should be restored.
4. Lead wires should be connected and laid as in the initial state.
5. Modification of the unit by user himself should absolutely be prohibited.
6. Tools and measuring instruments for use in repairs or inspection should be accurately calibrated in advance.
7. In installing the unit having been repaired, be careful to prevent the occurrence of any accident such as electrical shock, leak of current, or bodily injury due to the drop of any part.
8. To check the insulation of the unit, measure the insulation resistance between the power cord plug and grounding terminal of the unit. The insulation resistance should be $1\text{M}\Omega$ or more as measured by a 500V DC megger.
9. The initial location of installation such as window, floor or the other should be checked for being and safe enough to support the repaired unit again.
If it is found not so strong and safe, the unit should be installed at the initial location reinforced or at a new location.
10. Any inflammable thing should never be placed about the location of installation.
11. Check the grounding to see whether it is proper or not, and if it is found improper, connect the grounding terminal to the earth.



WORKING STANDARDS FOR PREVENTING BREAKAGE OF SEMICONDUCTORS

1. Scope

The standards provide for items to be generally observed in carrying and handling semiconductors in relative manufacturers during maintenance and handling thereof. (They apply the same to handling of abnormal goods such as rejected goods being returned).

2. Object parts

- (1) Micro computer
- (2) Integrated circuits (IC)
- (3) Field-effect transistors (FET)
- (4) P.C. boards or the like on which the parts mentioned in (1) and (2) of this paragraph are equipped.

3. Items to be observed in handling

- (1) Use a conductive container for carrying and storing of parts. (Even rejected goods should be handled in the same way).



Fig. 1. Conductive Container

- (2) When any part is handled uncovered (in counting, packing and the like), the handling person must always use himself as a body earth. (Make yourself a body earth by passing one M ohm earth resistance through a ring or bracelet).
- (3) Be careful not to touch the parts with your clothing when you hold a part even if a body earth is being taken.
- (4) Be sure to place a part on a metal plate with grounding.
- (5) Be careful not to fail to turn off power when you repair the printed circuit board. At the same time, try to repair the printed circuit board on a grounded metal plate.

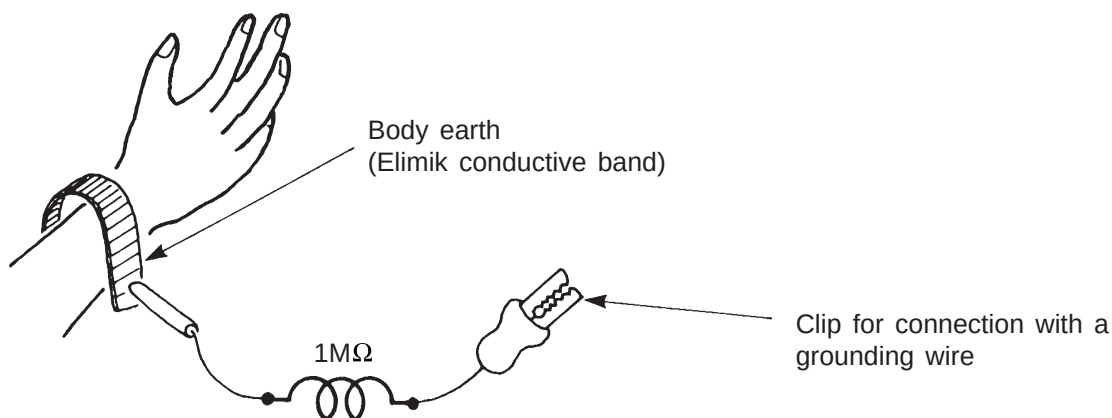


Fig. 2. Body Earth

- (6) Use a three wire type soldering iron including a grounding wire.

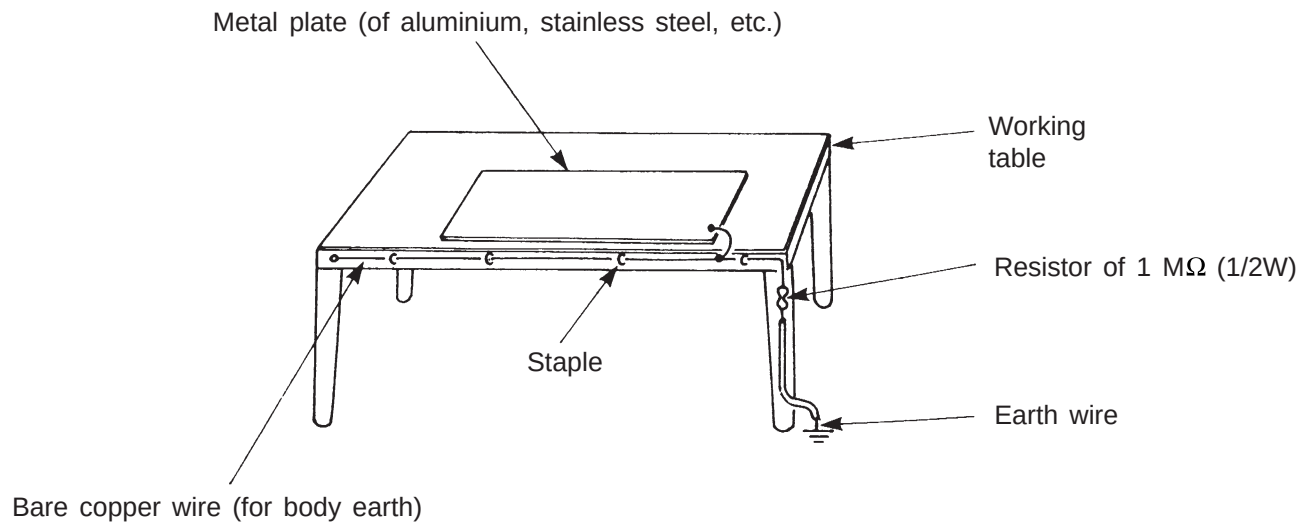


Fig. 3. Grounding of the working table

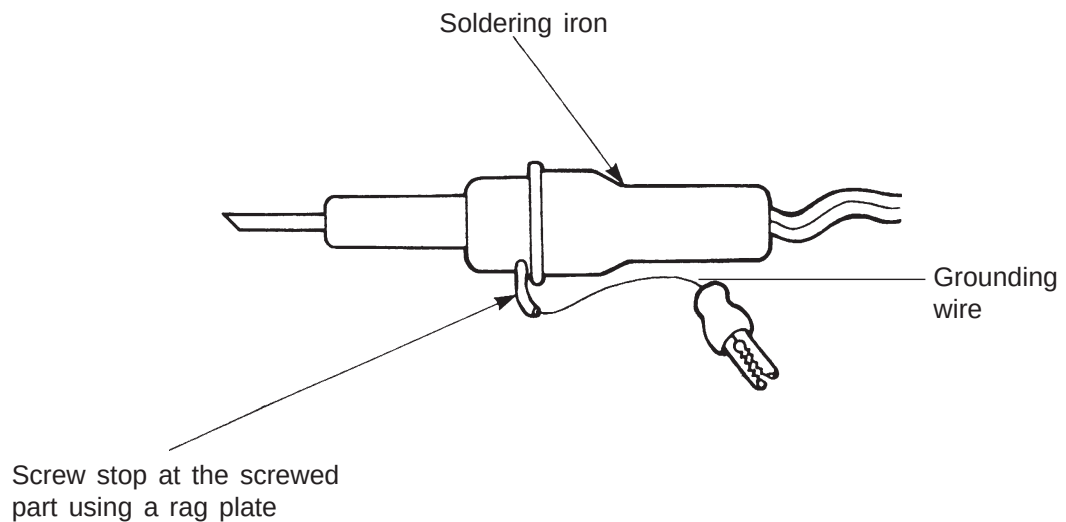


Fig. 4. Grounding a soldering iron

Use a high insulation mode (100V, 10MΩ or higher) when ordinary iron is to be used.

- (7) In checking circuits for maintenance, inspection or some others, be careful not to have the test probes of the measuring instrument shortcircuit a load circuit or the like.

⚠ CAUTION

1. In quiet operation or stopping the operation, slight flowing noise of refrigerant in the refrigerating cycle is heard occasionally, but this noise is not abnormal for the operation.
2. When it thunders near by, it is recommend to stop the operation and to disconnect the power cord plug from the power outlet for safety.
3. If the room air conditioner is stopped by setting the temperature or mis-operation, and then re-started in a moment, cooling operation does not start for 3 minutes, it is not abnormal and this is the result of the operation of IC delay circuit. This IC delay circuit ensures that there is no danger of blowing fuse or damaging parts even if operation is restarted accidentally.
4. This room air conditioner should not be used at the cooling operation when the outside temperature is below 20°C.
5. When the operation knob is set to“COOL ”from another position, IC delay circuit functions and stops the compressor for the first 3 minutes, which is not an abnormal phenomenon.

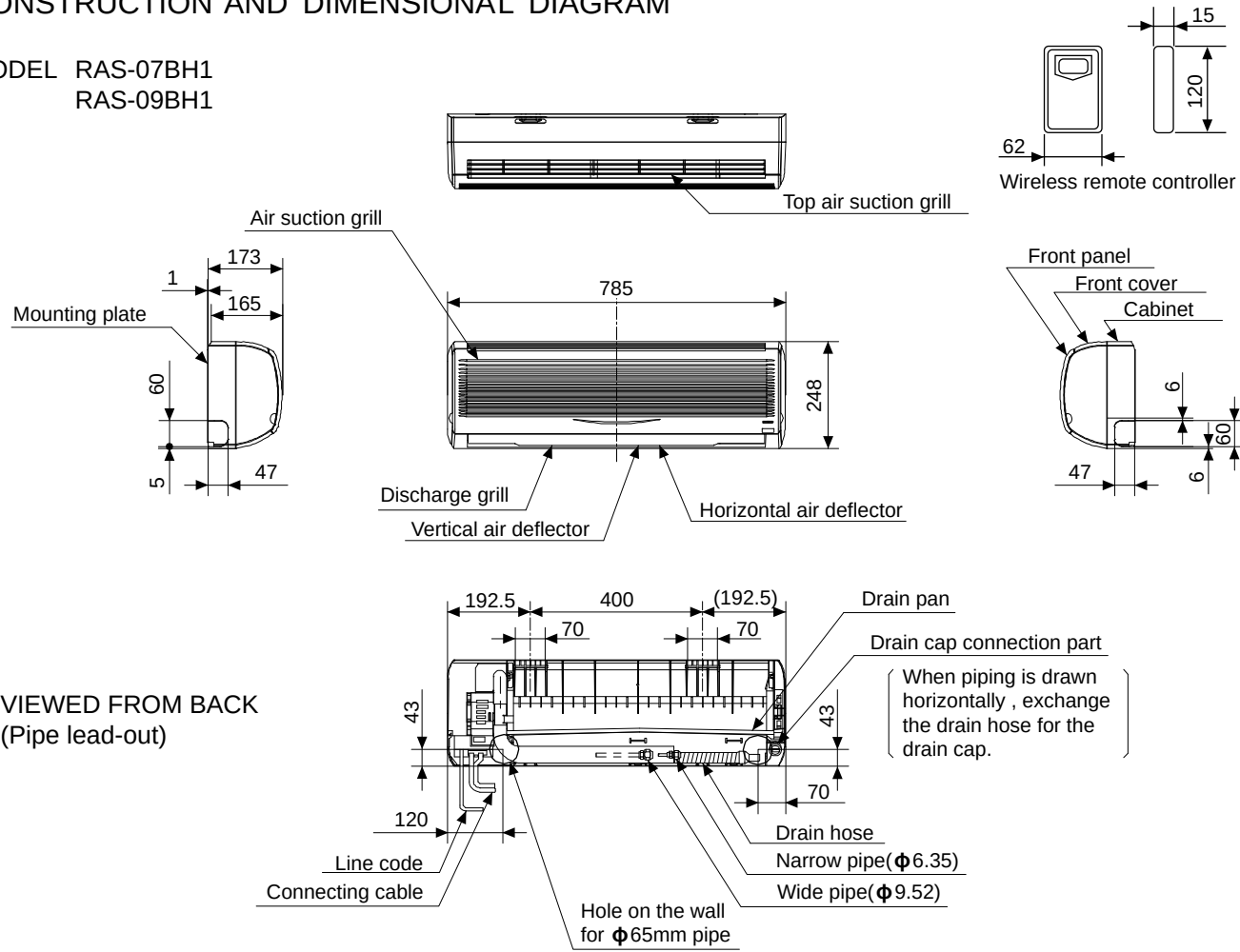
SPECIFICATIONS

MODEL		RAS-07BH1 RAS-09BH1	RAC-07BH1 RAC-09BH1	RAS-14BH1	RAC-14BH1
FAN MOTOR		8 W	20 W	10 W	20 W
FAN MOTOR CAPACITOR		1 μ F, 400 V	2.5 μ F, 400 V	1 μ F, 400 V	2.5 μ F, 400 V
FAN MOTOR PROTECTOR		YES	YES (INTERNAL)	YES	YES (INTERNAL)
COMPRESSOR		NO	SG533QB1	NO	SL232CV-C7LU
COMPRESSOR MOTOR CAPACITOR		NO	30 μ F, 400 V	NO	35 μ F, 400 V
OVERLOAD PROTECTOR		NO	YES	NO	YES
OVERHEAT PROTECTOR		NO	NO	NO	NO
FUSE (for MICRO COMPUTER)		3.15A	NO	3.15A	NO
POWER RELAY		891WP-A-C	NO	891WP-A-C	NO
POWER SWITCH		NO	NO	NO	NO
TEMPORARY SWITCH		YES	NO	YES	NO
SERVICE SWITCH		YES	NO	YES	NO
TRANSFORMER		YES	NO	YES	NO
VARISTOR		470NR	NO	470NR	NO
NOISE SUPPRESSOR		YES	NO	YES	NO
REMOTE CONTROL SWITCH (LIQUID CRYSTAL)		YES	NO	YES	NO
THERMOSTAT		YES (IC)	NO	YES (IC)	NO
FUSE CAPACITY		10 A TIME DELAY FUSE		15 A TIME DELAY FUSE	
REFRIGERANT CHARGING VOLUME (Refrigerant 22)	UNIT	-----	600g	-----	1,120g
	PIPES	WITHOUT REFRIGERANT BECAUSE COUPLING IS FLARE TYPE.			

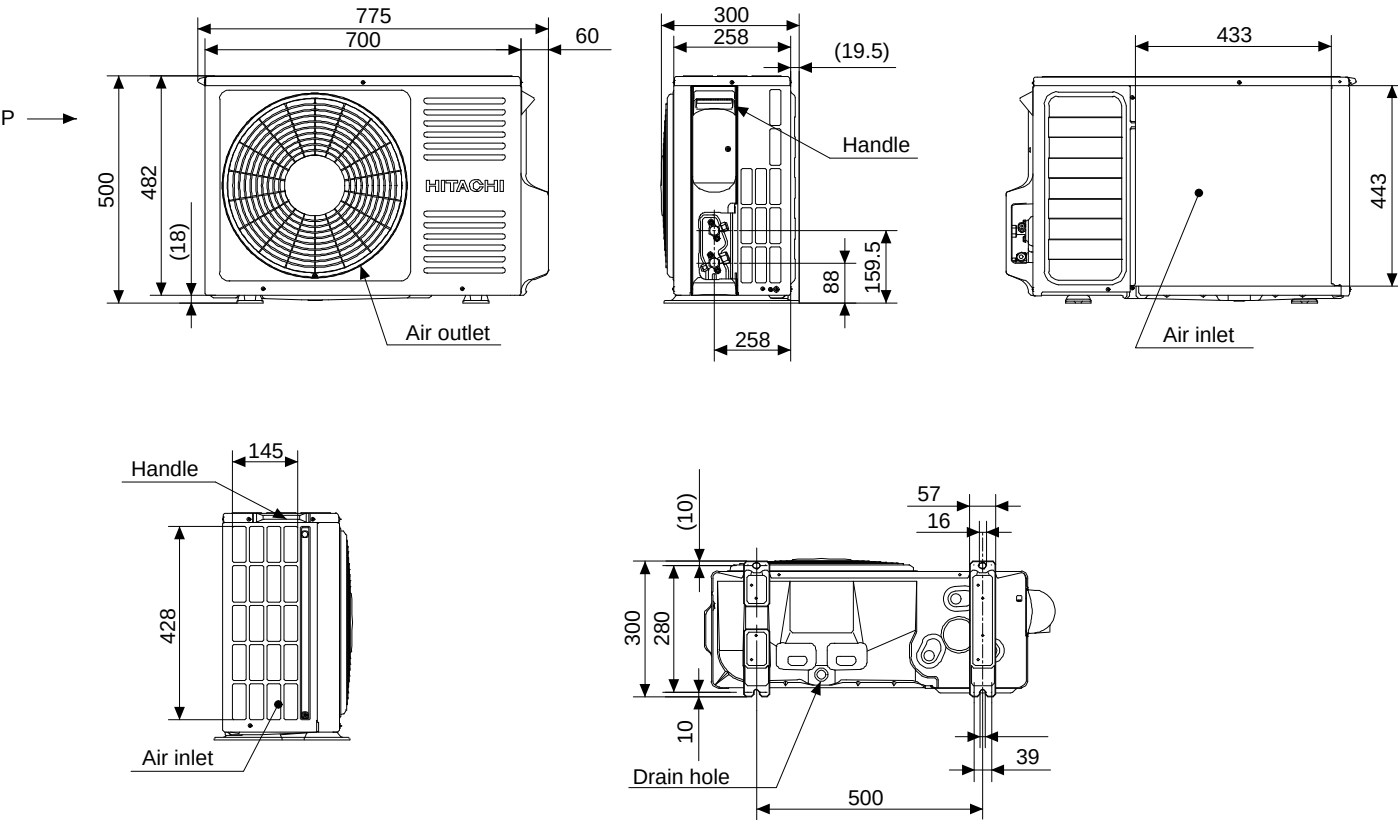
600g for piping set of 5m.
1,120g for piping set of 5m.

CONSTRUCTION AND DIMENSIONAL DIAGRAM

MODEL RAS-07BH1
RAS-09BH1

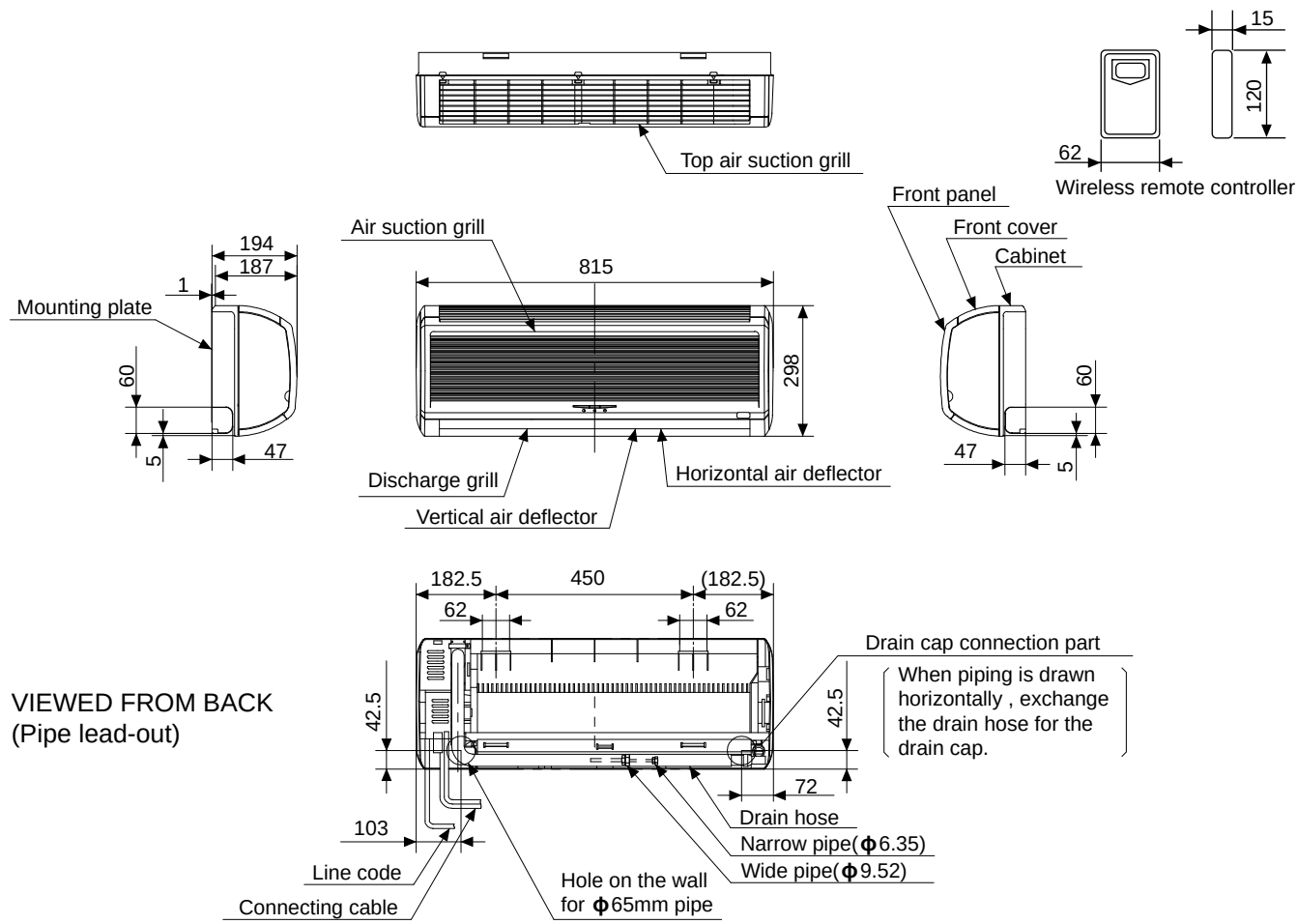


MODEL RAC-07BH1
RAC-09BH1

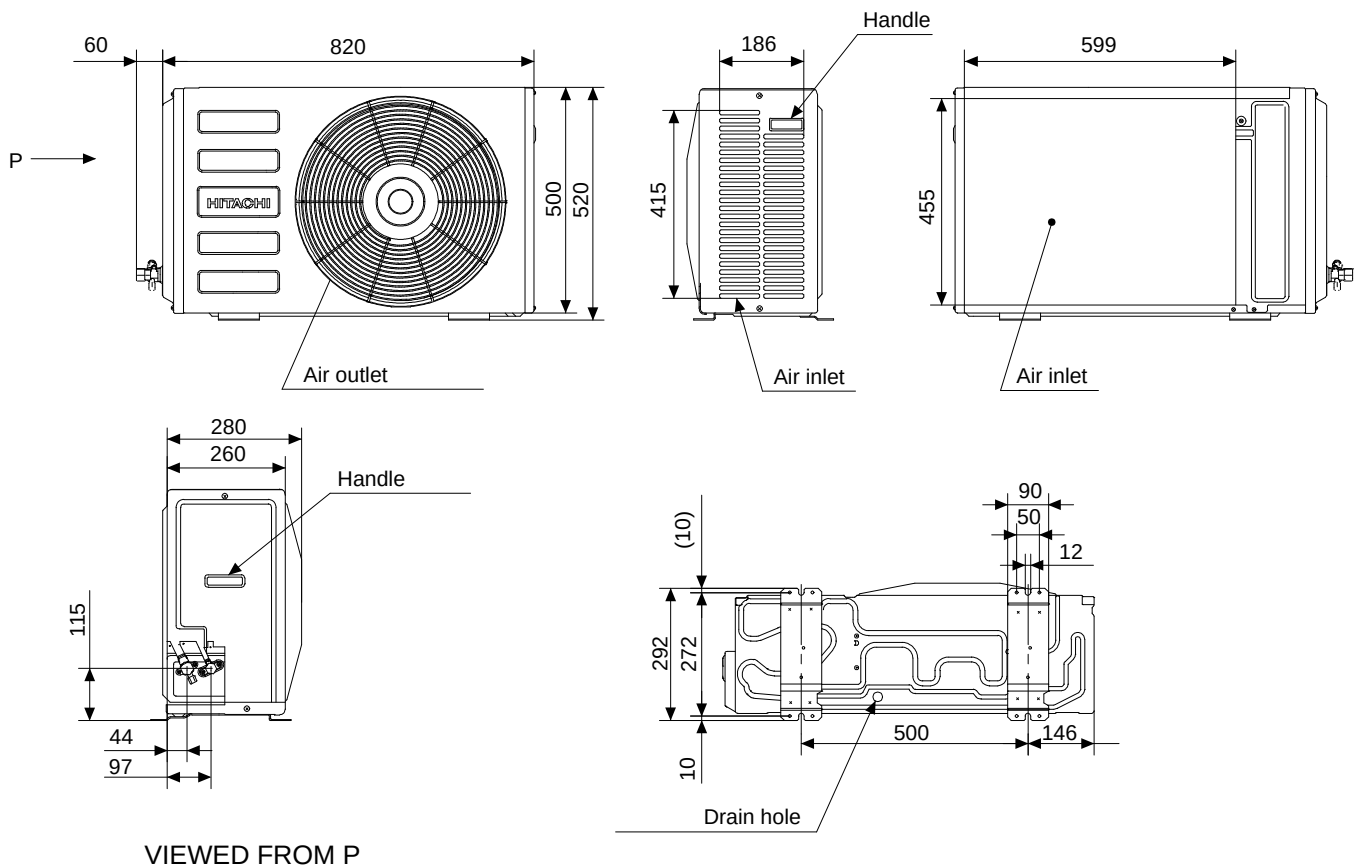


VIEWED FROM P

MODEL RAS-14BH1

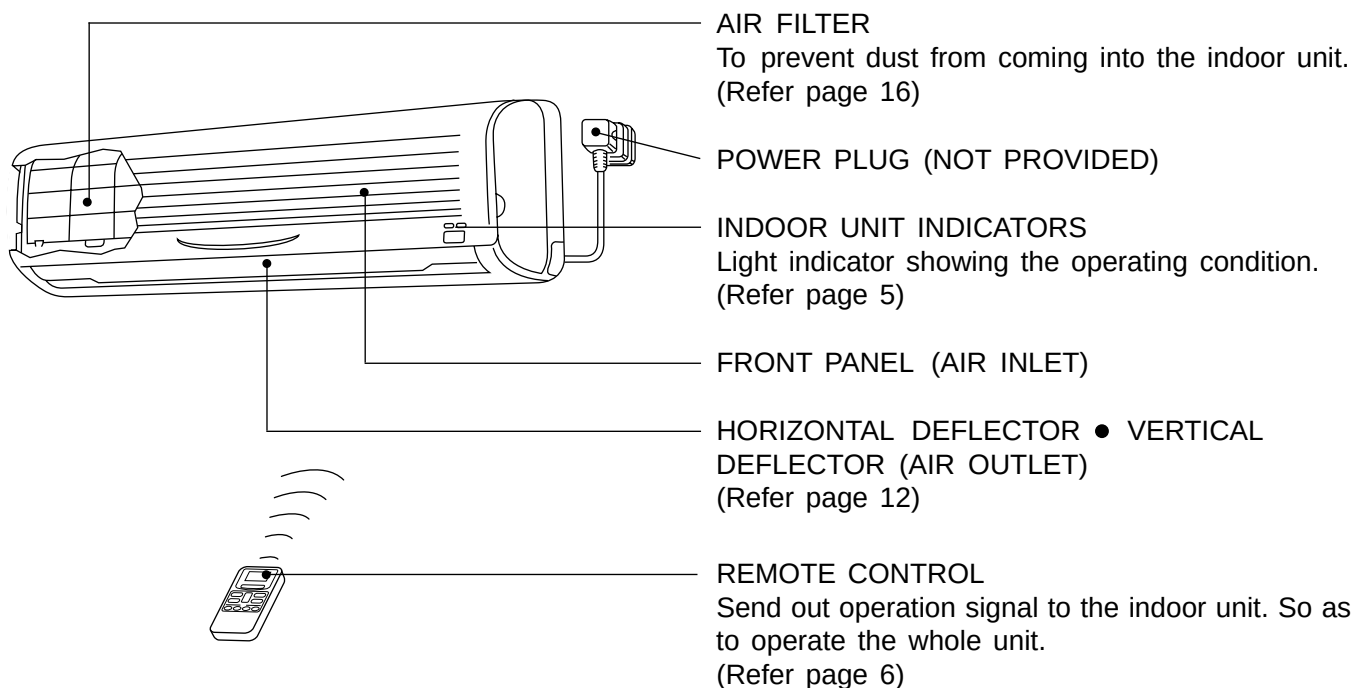


MODEL RAC-14BH1

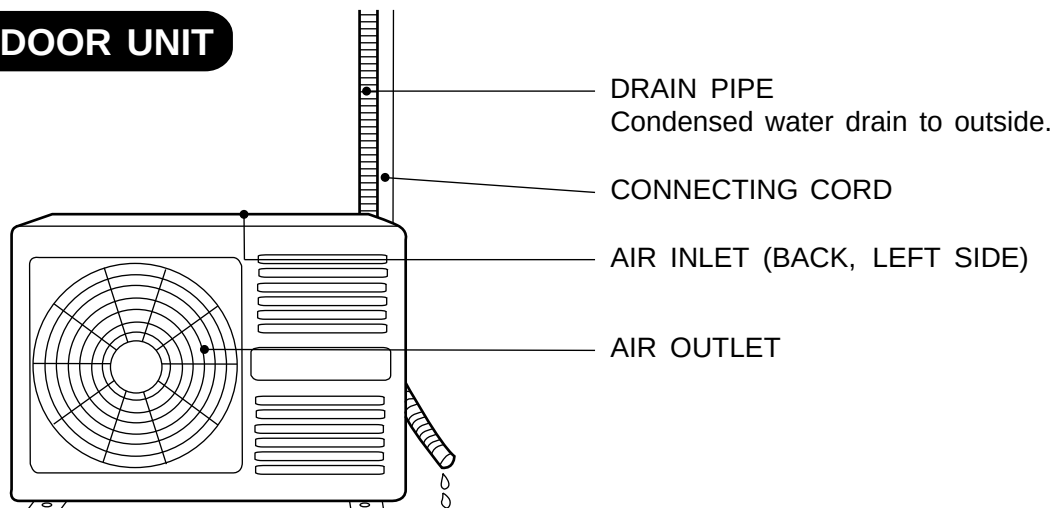


NAMES AND FUNCTIONS OF EACH PART

INDOOR UNIT



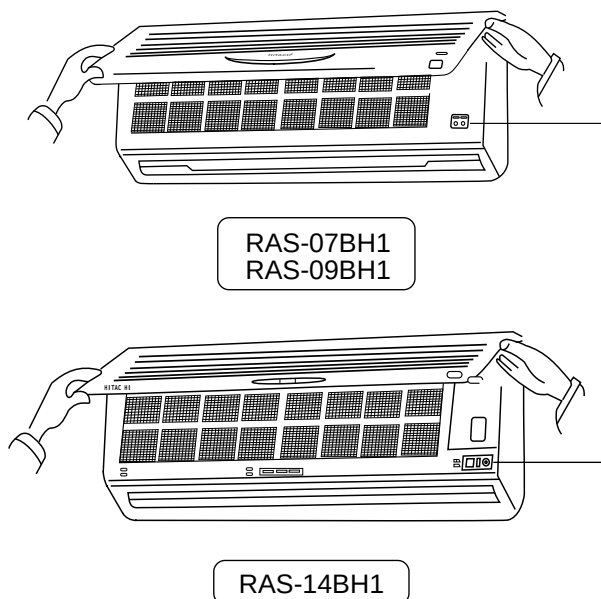
OUTDOOR UNIT



MODEL NAME AND DIMENSIONS

MODEL	WIDTH (mm)	HEIGHT (mm)	DEPTH (mm)
RAS-07BH1/RAS-09BH1	785	250	175
RAC-07BH1/RAC-09BH1	775	500	300
RAS-14BH1	815	298	194
RAC-14BH1	820	520	280

OPERATION INDICATOR

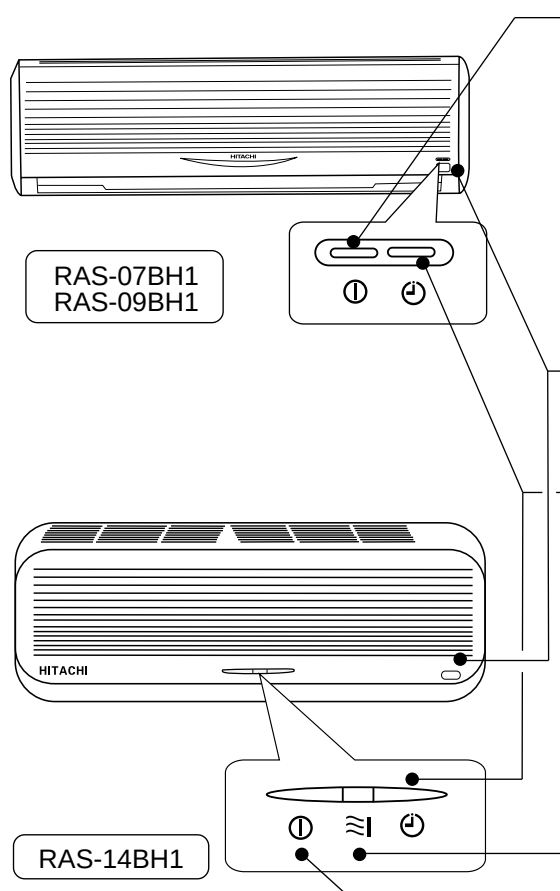


TEMPORARY SWITCH

Use this switch to start and stop when the remote controller does not work.

- By setting the temporary switch, the operation is done in previously set operation mode.
- When the operation is done using the temporary switch after the power source is turned off and turn on again, the operation is done in automatic mode.

INDOOR UNIT INDICATORS



OPERATION LAMP

This lamp lights during operation.

The OPERATION LAMP flashes in the following cases during heating.

(1) During preheating

For about 2–3 minutes after starting up.

(2) During defrosting

Defrosting will be performed about once an hour when frost forms on the heat exchanger of the outdoor unit, for 5–10 minutes each time.

SIGNAL RECEIVER

There will be a beep sound when this receiver receives signal from remote controller.

TIMER LAMP

This lamp lights when the timer is working.

HOT KEEP LAMP

This lamp lights in the following cases during heating. While this is lit, heated air will not come out of the indoor unit.

(1) During preheating

For about 2–3 minutes after starting up.

(2) During defrosting

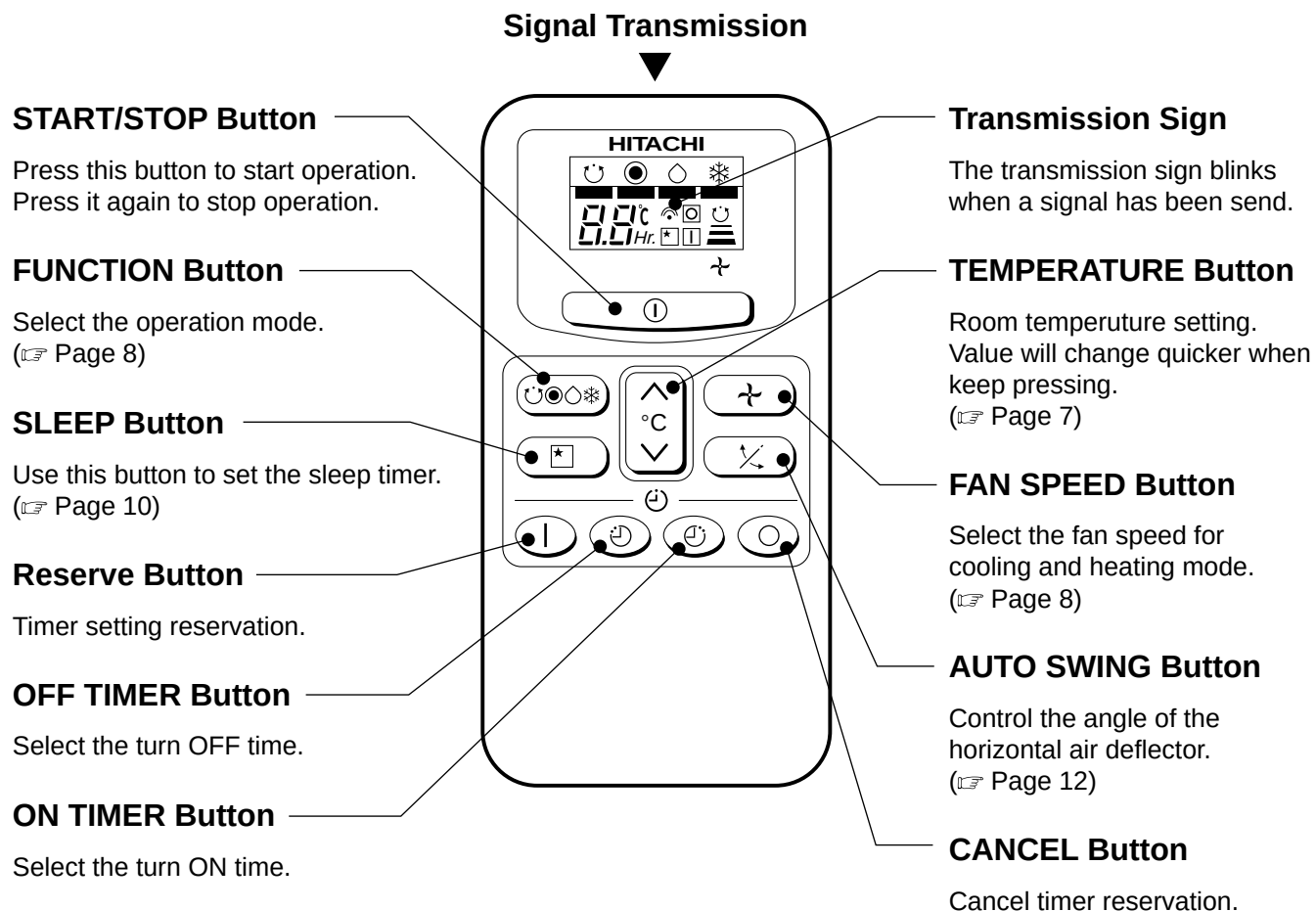
Defrosting will be performed about once an hour when frost forms on the heat exchanger of the outdoor unit, for 5–10 minutes each time.

OPERATION LAMP

This lamp lights during operation.

NAMES AND FUNCTIONS OF REMOTE CONTROL UNIT

- This controls the operation function and timer setting of the room air conditioner. The range of control is about 4 meters. If indoor lighting is controlled electronically, the range of control may be shorter.

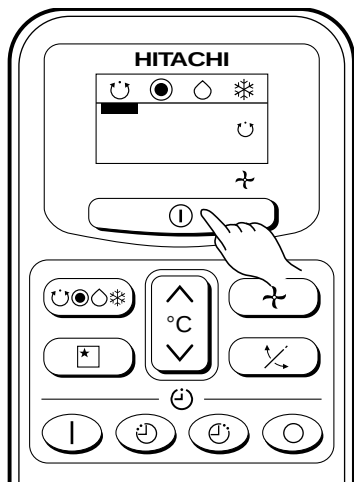


Precautions for Use

- Do not put the remote controller under direct sunlight and high temperature.
- Do not drop it on the floor, and protectit from water.
- If you press the FUNCTION button during operation, the air conditioner may stop for about 3 minutes for protection before you can start it again.

AUTOMATIC OPERATION

- The device will automatically determine the mode of operation, HEAT, COOL or DEHUMIDIFY, depending on the initial room temperature. The selected mode of operation will not change when the room temperature varies.



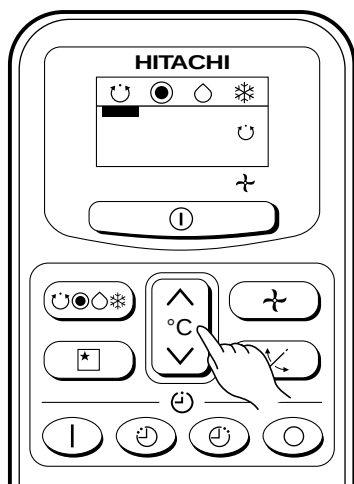
1 Press the (START/STOP) button

- Operation start with a “beep” sound.

STOP Press the (START/STOP) button

- Make sure the  (Automatic) mode have been selected by using the  (FUNCTION) button.
- The fan speed selector does not work at this operation.

- You can slightly adjust the room temperature.



1 Press the (ROOM TEMPERATURE) button

- Temperature setting change by 1°C for each 1 time press.



- You can raise or lower the temperature setting by a maximum of 3°C.
- The display does not indicate the preset temperature in the Automatic mode. Device will receive the setting by a “beep”.

■ CONDITION OF AUTOMATIC OPERATION

- The selected mode of operation will not change during the operation even though the room temperature change.

INITIAL ROOM TEMPERATURE (APPROX.)		FUNCTION	TEMPERATURE SETTING	FAN SPEED
Over 27°C	➡	COOL	27°C	HIGH at start, LOW after the preset temperature is reached
23 ~ 27°C	➡	DRY	Slightly lower than the room temperature	LOW
Under 23°C	➡	HEAT	23°C	HIGH at start, MED or LOW after the preset temperature is reached

MANUAL OPERATION [Heating • Dehumidify • Cooling]

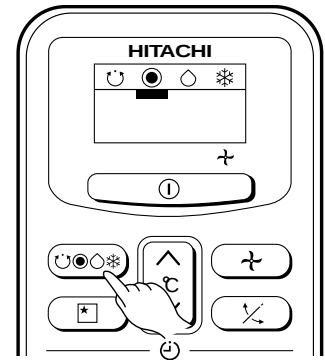
- Please use under below condition when you want to set the function mode, room temperature and fan speed by manually.

HEATING	DEHUMIDIFYING	COOLING
Outdoor Temperature 24°C below	Room Temperature 16°C above	Outdoor Temperature 21°C above

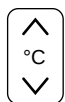
1 OPERATION MODE SELECTION

- Every time you press the button, the mode will change as the below sequence

☺ (Auto) → ● (Heat) → ○ (Dry) → ❄ (Cool)



2 ROOM TEMPERATURE SETTING

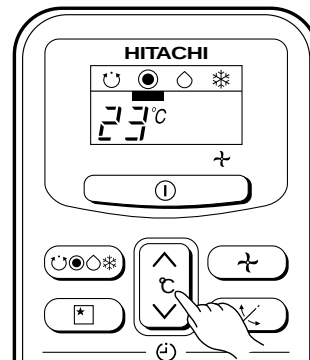


◀ Up
◀ Down

■ Recommend Temperature

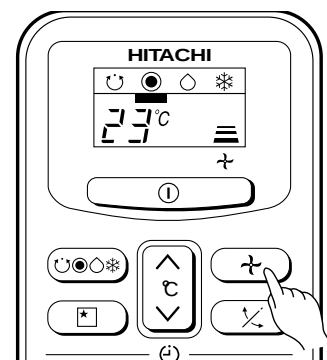
Heating	20~24°C
Dehumidify	20~26°C
Cooling	25~28°C

- The cooling operation does not start if the temperature setting is higher than the current room temperature.



3 FAN SPEED SETTING

- Every time you press the button, fan speed will change as below sequence.
- **HEATING** } : AUTO → HIGH → MED → LOW
- **COOLING** } : (☺ → ≡ → = → —)
- **DEHUMIDIFYING** : LOW (FIXED)
- : (—)

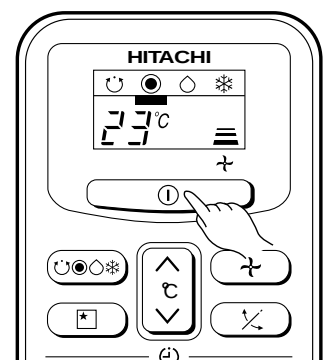


4 Press the ① (START/STOP) button

- Operation start with a signal received sound "beep".

STOP Press the ① (START/STOP) button

- As the settings are stored in memory in the remote control unit, you only have to press the ① (Start/Stop) button in order to use the same setting next time.



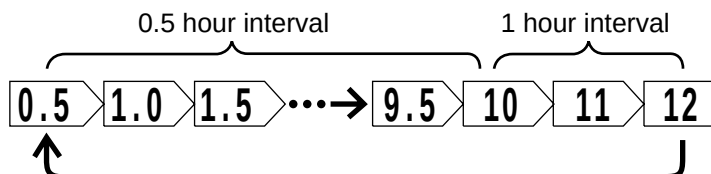
TIMER RESERVATION

■ ON Timer and OFF Timer are available.

OFF Timer Reservation

1 OFF TIME setting

- Select the OFF TIME by pressing the  (OFF) Button.
- Setting time will change according to the below sequence when you press the button.



- The value change quicker if you keep pressing the button.

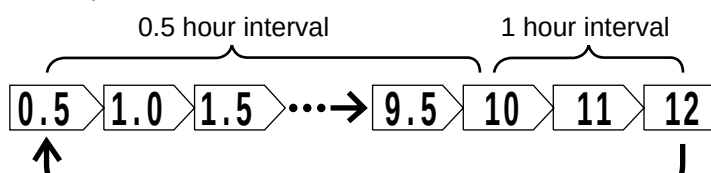
2 Press the (Reserve) button

- OFF TIMER reserved with a signal received sound “beep”.
- The  (OFF) Mark starts lighting instead of blinding.

ON Timer Reservation

1 ON TIME setting

- Select the ON TIMER by pressing the  (ON) Button.
- At the beginning of setting, time 6 hours was set.
- Setting time will change according to the below sequence.



- The value change quicker if you keep pressing the button.

2 Press the (Reserve) button

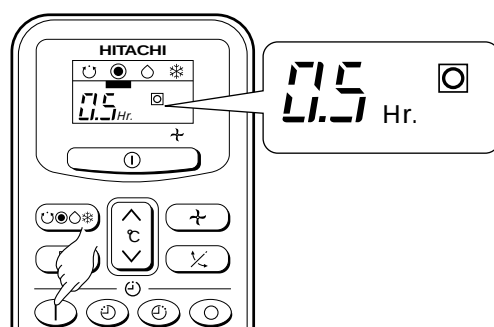
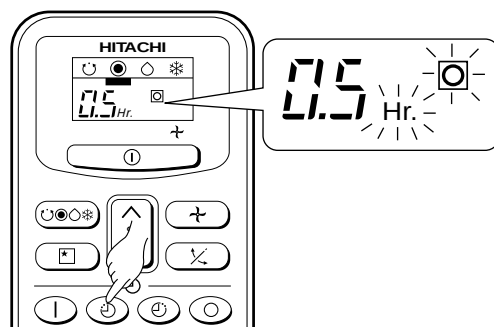
- ON TIMER reserved with a signal received sound “beep”.
- The  (ON) Mark starts lighting instead of blinding.

CANCELLATION of Timer Reservation

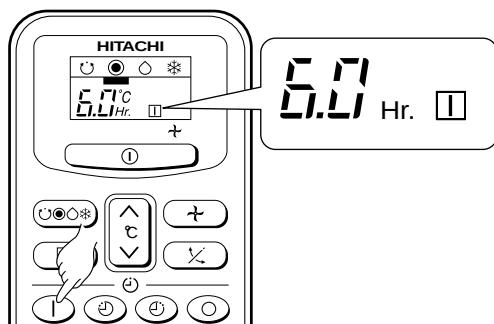
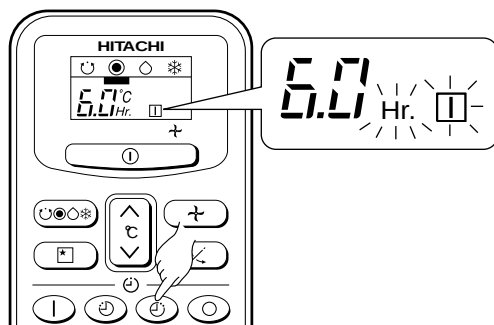
1 Press the (Cancel) button

- As the time settings are stored in remote controller memory, you only have to press the  (Reserve) button in order to use the same setting next time.

■ Operation stop at setting time

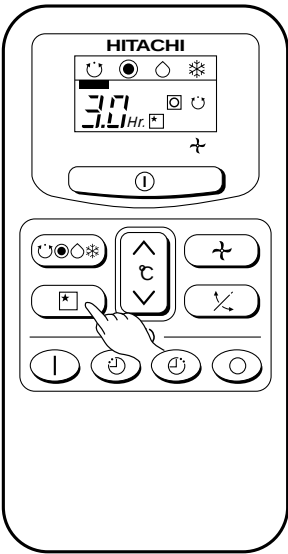


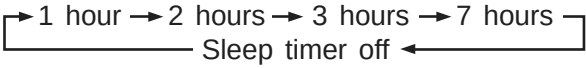
■ Operation will start for setting temperature at setting time (The starting time may different depend on the room temperature and set temperature).



HOW TO SET THE SLEEP TIMER

Press the  (SLEEP) button, and the display changes as shown below.



Mode	Indication
Sleep timer	

Sleep Timer: The device will continue working for the designated number of hours and then turn off.
Point the signal window of the remote controller toward the indoor unit, and press the SLEEP button.
The timer information will be displayed on the remote controller.
The TIMER lamp lights with a beep from the indoor unit.

Example: Setting 3 hours sleep time.

How to Cancel Reservation

Point the signal window of the remote controller toward the indoor unit, and press the  (CANCEL) button.

The  (RESERVED) sign goes out with a beep and the  (TIMER) lamp turns off on the indoor unit.

NOTE

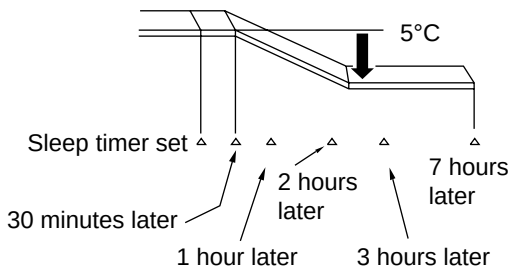
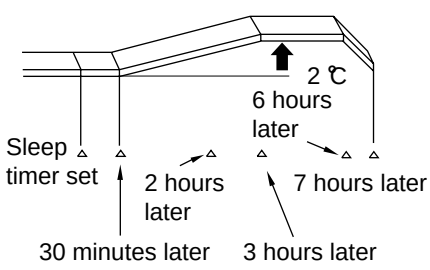
If you set the sleep timer when the off-timer or on/off-timer has been set earlier, the sleep timer becomes effective instead of the off- or on/off-timer set earlier.

Explanation of the sleep timer

The device will control the FAN SPEED and room temperature automatically so as to be quiet and good for people's health.

You can set the sleep timer to turn off after 1, 2, 3 or 7 hours. The FAN SPEED and room temperature will be controlled as shown below.

Operation with the sleep timer

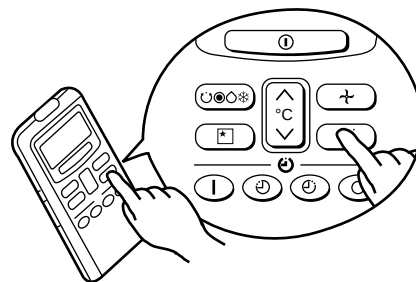
Function	Operation	
Heating “ ● ”	The room temperature will be controlled 5°C below the temperature setting 30 minutes after the setting of the sleep timer. The FAN SPEED will be set to LOW immediately.	The room temperature is kept at about 12°C minimum.  The diagram shows a line graph of temperature over time. The temperature starts at a baseline, then drops 5°C 30 minutes after the sleep timer is set. It remains at this lower level until 7 hours later, when it returns to the baseline. The graph is labeled with 'Sleep timer set', '30 minutes later', '1 hour later', '2 hours later', '3 hours later', and '7 hours later'. A downward arrow indicates the 5°C drop.
Cooling “ ❄ ” and dehumidifying “ ◐ ”	The FAN SPEED will be set to LOW immediately.	The room temperature is kept at about 25-28°C.  The diagram shows a line graph of temperature over time. The temperature starts at a baseline, then rises 2°C 30 minutes after the sleep timer is set. It remains at this higher level until 7 hours later, when it returns to the baseline. The graph is labeled with 'Sleep timer set', '30 minutes later', '2 hours later', '3 hours later', '6 hours later', and '7 hours later'. An upward arrow indicates the 2°C rise.
Fan “ ✚ ”	The settings of room temperature and circulation are not varied.	

ADJUSTING THE AIR DEFLECTOR

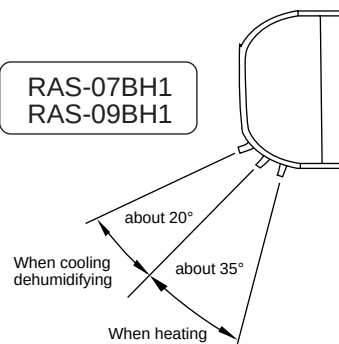
1

Adjustment of the conditioned air in the upward and downward directions.

According to operation, the horizontal air deflector is automatically set to the proper angle suitable for each operation. The deflector can be swing up and down and also set to the desired angle using the “ (AUTO SWING)” button. (If the angle of the deflector is changed, it will not return to the auto-set position after operations start unless the operation mode is switched.)

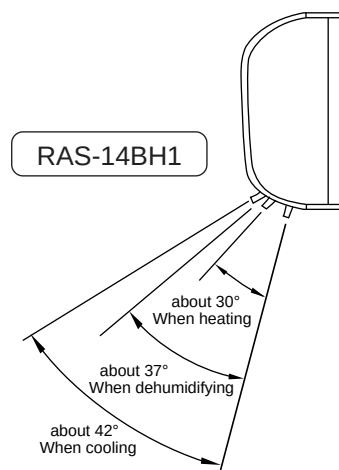


- If the “ (AUTO SWING)” button is pressed once, the horizontal air deflector swings up and down. If the button is pressed again, the deflector stops in its current position. Several seconds (about 6 seconds) may be required before the deflector starts to move.
- Use the horizontal air deflector within the adjusting range shown on the right.
- When the operation is stopped, the horizontal air deflector moves and stops at the position where the air outlet closes.



CAUTION

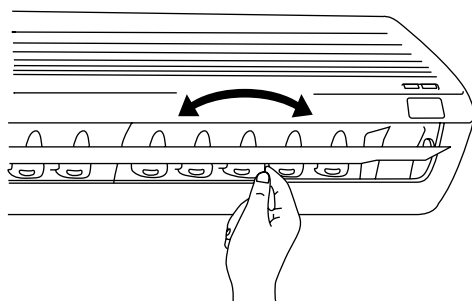
- In “Cooling” operation, do not keep the horizontal air deflector swinging for a long time. Some dew may form on the horizontal air deflector and some dew may drop from it.



2

Adjustment of the conditioned air to the left and right.

Hold the vertical air deflector as shown in the figure and adjust the conditioned air to the left and right.

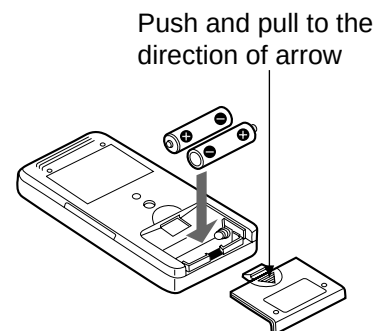


HOW TO EXCHANGE THE BATTERIES IN THE REMOTE CONTROLLER

- 1 Remove the cover as shown in the figure and take out the old batteries.



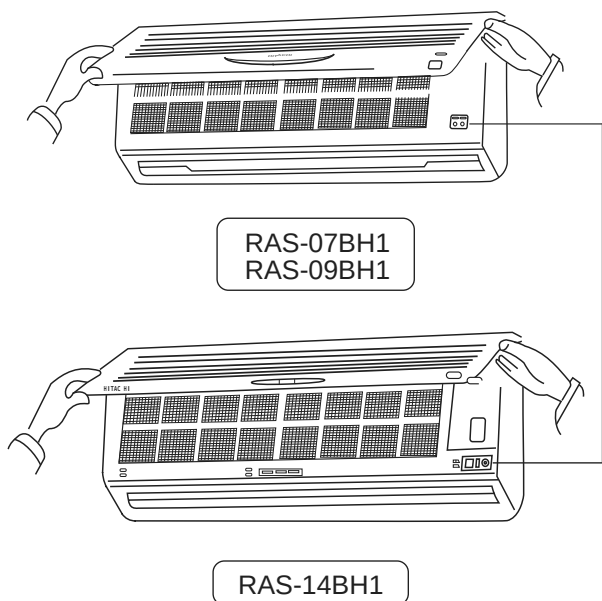
- 2 Install the new batteries.
The direction of the batteries should match the marks in the case.



⚠ CAUTION

1. Do not use new and old batteries, or different kinds of batteries together.
2. Take out the batteries when you do not use the remote controller for 2 or 3 months.

TEMPORARY SWITCH



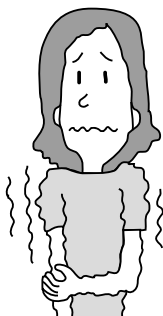
TEMPORARY SWITCH

Use this switch to start and stop when the remote controller does not work.

- By setting the temporary switch, the operation is done in previously set operation mode.
- When the operation is done using the temporary switch after the power source is turned off and turn on again, the operation is done in automatic mode.

THE IDEAL WAYS OF OPERATION

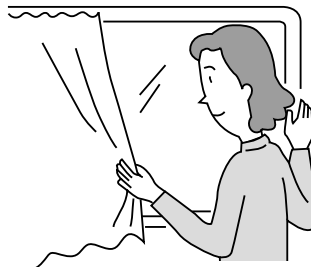
Suitable Room Temperature



⚠ Warning

Freezing temperature is bad for health and a waste of electric power.

Install curtain or blinds

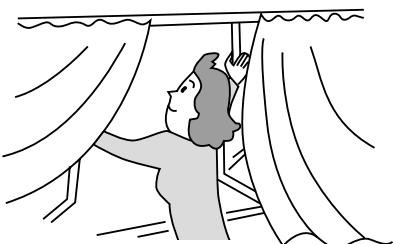


It is possible to reduce heat entering the room through windows.

Ventilation

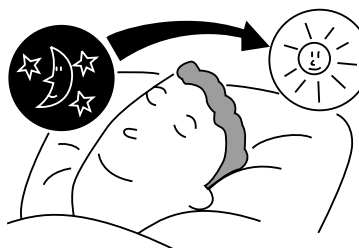
⚠ Caution

Do not close the room for a long period of time. Occasionally open the door and windows to allow the entrance of fresh air.



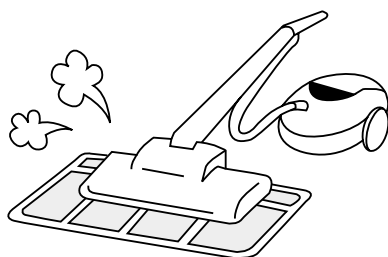
Effective Usage Of Timer

At night, please use the "OFF or ON timer operation mode", together with your wake up time in the morning. This will enable you to enjoy a comfortable room temperature. Please use the timer effectively.



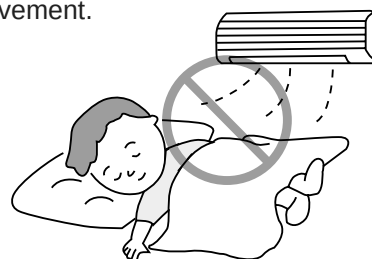
Do Not Forget To Clean The Air Filter

Dusty air filter will reduce the air volume and the cooling efficiency. To prevent from wasting electric energy, please clean the filter every 2 weeks.



Please Adjust Suitable Temperature For Baby And Children

Please pay attention to the room temperature and air flow direction when operating the unit for baby, children and old folks who have difficulty in movement.

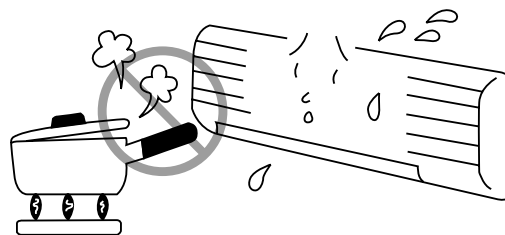


FOR USER'S INFORMATION

The Air Conditioner And The Heat Source In The Room

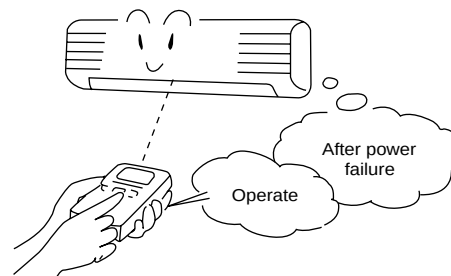
⚠ Caution

If the amount of heat in the room is above the cooling capability of the air conditioner (for example: more people entering the room, using heating equipments and etc.), the preset room temperature cannot be achieved.



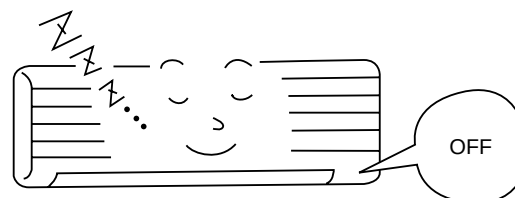
After Power Failure

When the power is resumed after a power failure, the indoor unit will still remain "OFF". To operate the unit, please press the "ON/OFF" button again.



Not Operating For A Long Time

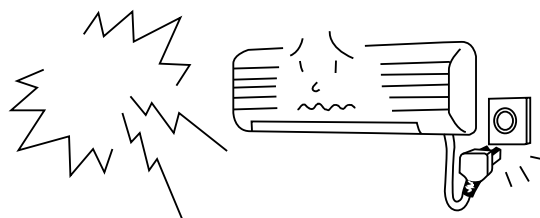
When the indoor unit is not to be used for a long period of time, please switch off the power from the mains. If the power from mains remains "ON", the indoor unit still consumes about 8W in the operation control circuit even if it is in "OFF" mode.



When Lightning Occurs

⚠ Warning

To protect the whole unit during lightning, please stop operating the unit and remove the plug from the socket.



MAINTENANCE

Before the cleaning, stop operation and disconnect the power supply.

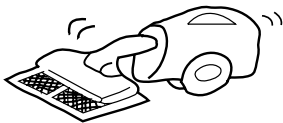
1. AIR FILTER

Clean the air filter, as it removes dust inside the room. In case the air filter is full of dust, the air flow will decrease and the cooling capacity will be reduced. Further, noise may occur. Be sure to clean the filter following the procedure below.

PROCEDURE

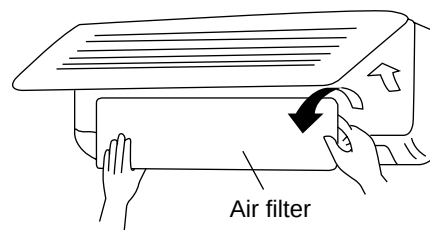
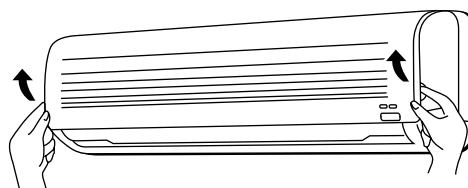
- 1 Open the front panel and remove the filter
 - Gently lift and remove the air cleansing and deodorizing filters from the air filter frame.

- 2
 - Vacuum dust from the air filter using vacuum cleaner. If there is too much dust, rinse under running tap water and gently brush it with soft bristle brush. Allow filters to dry in shade.

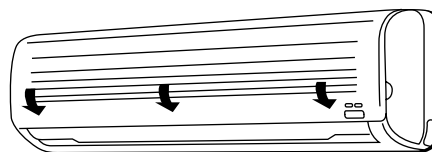
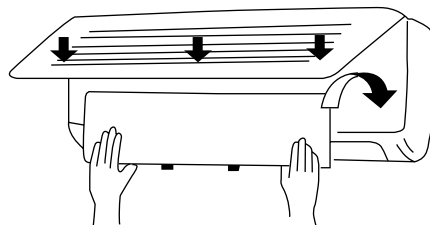


- 3
 - Set the filter with "FRONT" mark facing front, and slot them into the original state.
 - After attaching the filter, push the front panel at three arrow portions as shown in figure and close it.

REMOVING METHOD



INSTALLATION METHOD



CAUTION

- Do not wash with hot water at more than 40°C. The filter may shrink.
- When washing it, shake off moisture completely and dry it in the shade; do not expose it directly to the sun. The filter may shrink.

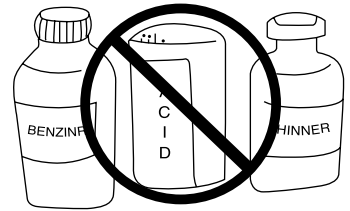
2. Washable Front Panel

- The front panel remove and can wash with clean water in whole.
Wash it with a soft sponge.
After using neutral detergent, wash thoroughly with clean water.
- When it is not removed, wipe it with a soft dry cloth. Wipe the remote controller thoroughly with a soft dry cloth.
- Wipe the water thoroughly.
If the water remains at indicators or signal receiver of indoor unit, it causes trouble.



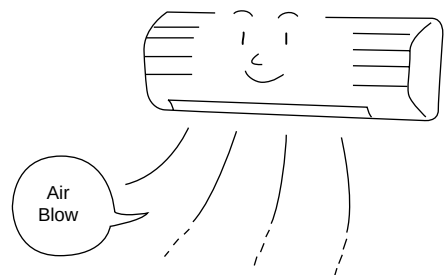
⚠ CAUTION

- Do not splash or direct water to the body of the unit when cleaning it as this may cause short circuit.
- Never use hot water (above 40°C), benzine, gasoline, acid, thinner or a brush, because they will damage the plastic surface and the coating.



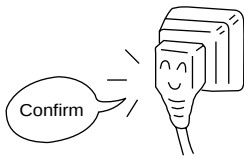
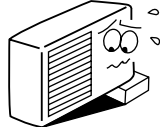
3. MAINTENANCE AT BEGINNING OF LONG OFF PERIOD

- Running the unit setting the operation mode to ❄ (COOL), the temperature to 32°C and the fan speed to HI for about half a day on a fine day, and dry the whole of the unit.
- Disconnect the power plug.



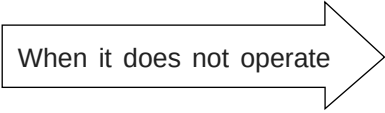
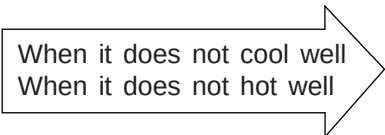
REGULAR INSPECTION

PLEASE CHECK THE FOLLOWING POINTS EVERY EITHER HALF YEARLY OR YEARLY. CONTACT YOUR SALES AGENT SHOULD YOU NEED ANY HELP.

1		Is the plug of power line firmly plugged into the socket? (Please ensure no loose contact between them).
2		Is the earth line disconnected or broken?
3		Is the mounting frame seriously affected by rust and is the outdoor unit tilted or unstable?

AFTER SALE SERVICE AND WARRANTY

WHEN ASKING FOR SERVICE, CHECK THE FOLLOWING POINTS.

CONDITION	CHECK THE FOLLOWING POINTS
 When it does not operate	● Is the fuse all right? ● Is the voltage extremely high or low? ● Is the circuit breaker "ON"?
 When it does not cool well When it does not hot well	● Is the air filter blocked with dust? ● Does sunlight fall directly on the outdoor unit? ● Is the air flow of the outdoor unit obstructed? ● Are the doors or windows opened, or is there any source of heat in the room? ● Is the set temperature suitable?



Notes

- In quiet operation or stopping the operation, the following phenomena may occasionally occur, but they are not abnormal for the operation.
 - (1) Slight flowing noise of refrigerant in the refrigerating cycle.
 - (2) Slight rubbing noise from the fan casing which is cooled and then gradually warmed as operation stops.
- The odor will possibly be emitted from the room air conditioner because the various odor, emitted by smoke, foodstuffs, cosmetics and so on, sticks to it. So please clean the air filter and the evaporator regularly to reduce the odor.

- Please contact your sales agent immediately if the air conditioner still fails to operate normally after the above inspections. Inform your agent of the model of your unit, production number, date of installation. Please also inform him regarding the fault.
- Power supply shall be connected at the rated voltage, otherwise the unit will be broken or could not reach the specified capacity.

MAIN PARTS COMPONENT

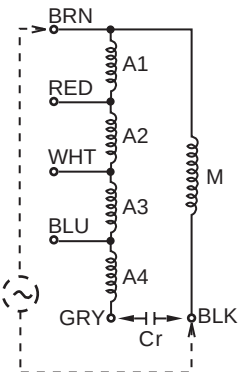
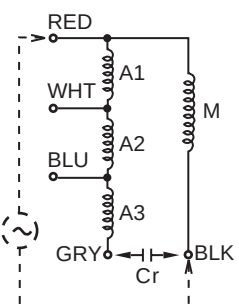
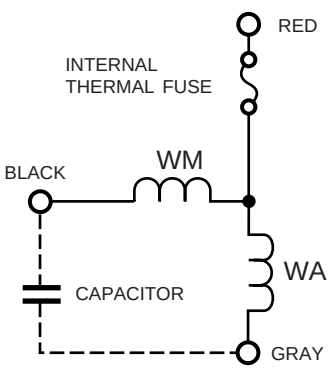
THERMOSTAT

Thermostat Specifications

MODEL			RAS-07BH1 RAS-09BH1 RAS-14BH1	RAS-07BH1 RAS-09BH1	RAS-14BH1
THERMOSTAT MODEL			IC		
OPERATION MODE			COOL	HEAT	
TEMPERATURE °C (°F)	INDICATION 16	ON	17.0 (62.6)	15.7 (60.3)	16.7 (62.1)
		OFF	16.3 (61.3)	16.3 (61.3)	17.3 (63.1)
	INDICATION 24	ON	25.0 (77.0)	23.7 (74.7)	24.7 (76.5)
		OFF	24.3 (75.7)	24.3 (75.7)	25.3 (77.5)
	INDICATION 32	ON	33.0 (91.4)	31.7 (89.1)	32.7 (90.9)
		OFF	30.6 (90.1)	33.3 (90.1)	33.3 (91.9)

FAN MOTOR

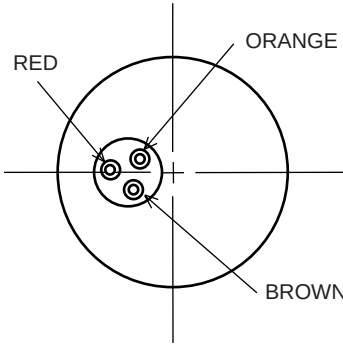
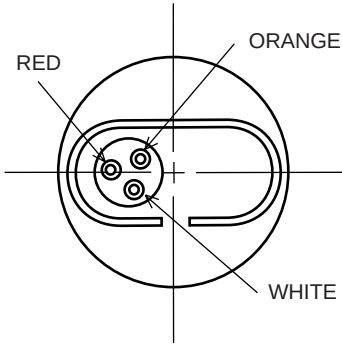
Fan Motor Specifications

MODEL		RAS-07BH1 RAS-09BH1	RAS-14BH1	RAC-07BH1 RAC-09BH1	RAC-14BH1
PHASE		SINGLE			
RATED VOLTAGE		220 V			
RATED FREQUENCY		50 Hz			
OUTPUT		10 W	8 W	20 W	
POLE NUMBER		4			6
CONNECTION					
RESISTANCE VALUE (Ω)	20°C (68°F)	M = 300.0 A1 = 107.0 A2 = 112.0 A3 = 90.0 A4 = 118.0	M = 481.0 A1 = 73.0 A2 = 84.0 A3 = 328.0	WA = 217.0 WM = 347.0	WA = 239.0 WM = 378.0

COMPRESSOR MOTOR

Compressor Motor Specifications

MODEL		RAC-07BH1, RAC-09BH1	RAC-14BH1
COMPRESSOR MODEL		SG533QB1	SL232CV-C7LU
PHASE		SINGLE	SINGLE
RATED VOLTAGE		220 V	220 V
RATED FREQUENCY		50 Hz	50 Hz
LOCKED ROTOR CURRENT		18 A	27 A
POLE NUMBER		2	2
CONNECTION			
RESISTANCE VALUE (Ω)	20°C (68°F)	RA = 3.76 RM = 3.73	RA = 2.95 RM = 4.04



CAUTION

When the refrigerating cycle has been operated for a long time with the capillary tubes clogged or crushed or with too little refrigerant, check the color of the refrigerating machine oil inside the compressor. If the color has been changed conspicuously, replace the compressor.

WIRING DIAGRAM

MODEL RAS-07BH1 / RAC-07BH1
RAS-09BH1 / RAC-09BH1
RAS-14BH1 / RAC-14BH1

(A) : COMPRESSOR

(B) : FAN MOTOR

(C) : FAN MOTOR PROTECTOR

(D) : 1 μ F CAPACITOR

(E) : COMPRESSOR CAPACITOR

RAC-07BH1, RAC-09BH1 : 30 μ F

RAC-14BH1 : 35 μ F

(H) : 2.5 μ F CAPACITOR

(I) : FAN MOTOR PROTECTOR (INTERNAL)

(J) : LINE CORD

(K) : TERMINAL BOARD

(L) : OUTDOOR FAN RELAY

(M) : THERMAL FUSE FOR PCB

(N) : REVERSING VALVE COIL

(P) : POWER RELAY

(R) : SURGE ASSORBER

(S) : THERMISTOR

(T) : TRANSFORMER

(U)_P : OVER LOAD PROTECTOR (INTERNAL)

(V) : VARISTOR

(W) : INDOOR FAN RELAY

(X) : FUSE

(Y) : REVERSING VALVE RELAY

(Z) : AUTO SWEEP MOTOR

BLU : BLUE

YEL : YELLOW

BRN : BROWN

WHT : WHITE

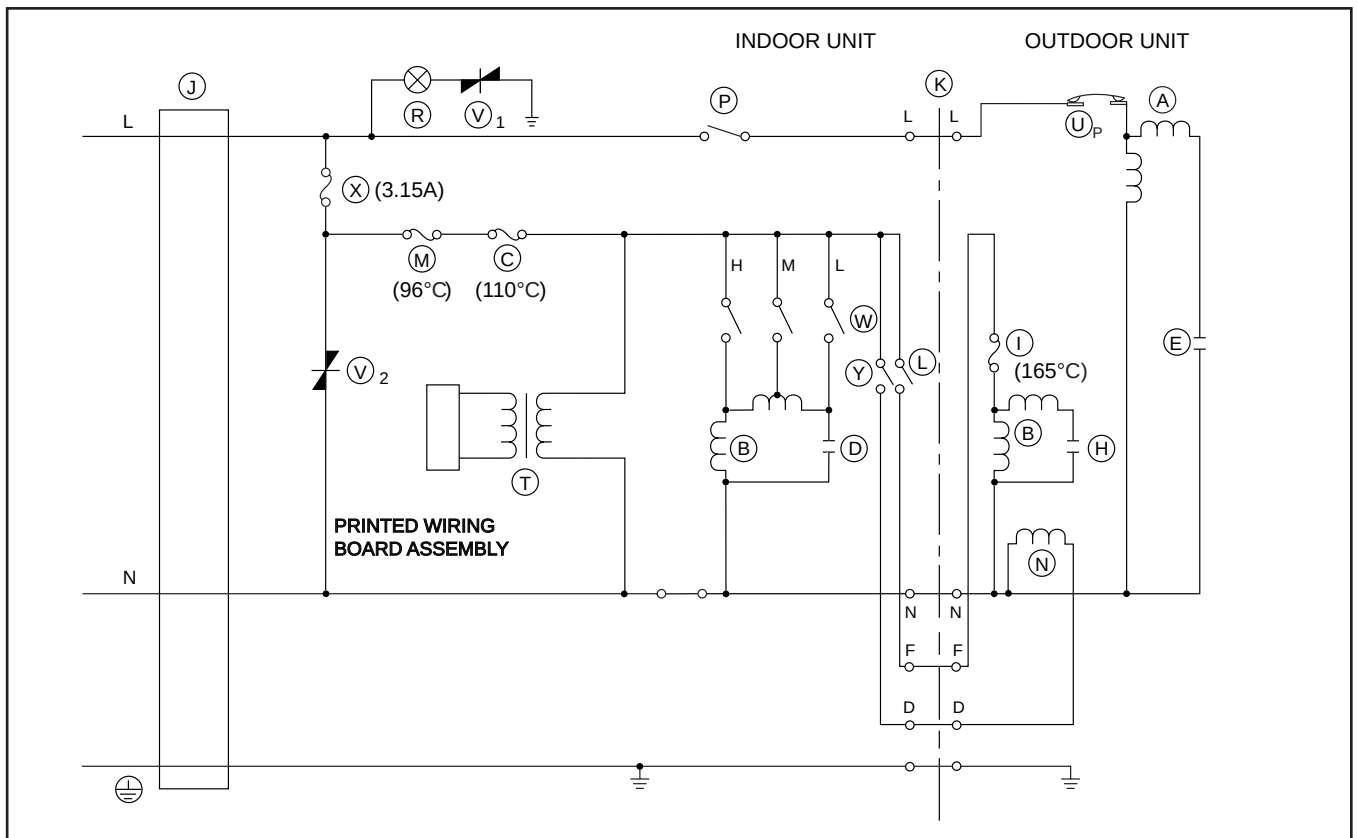
GRY : GRAY

ORN : ORANGE

GRN : GREEN

RED : RED

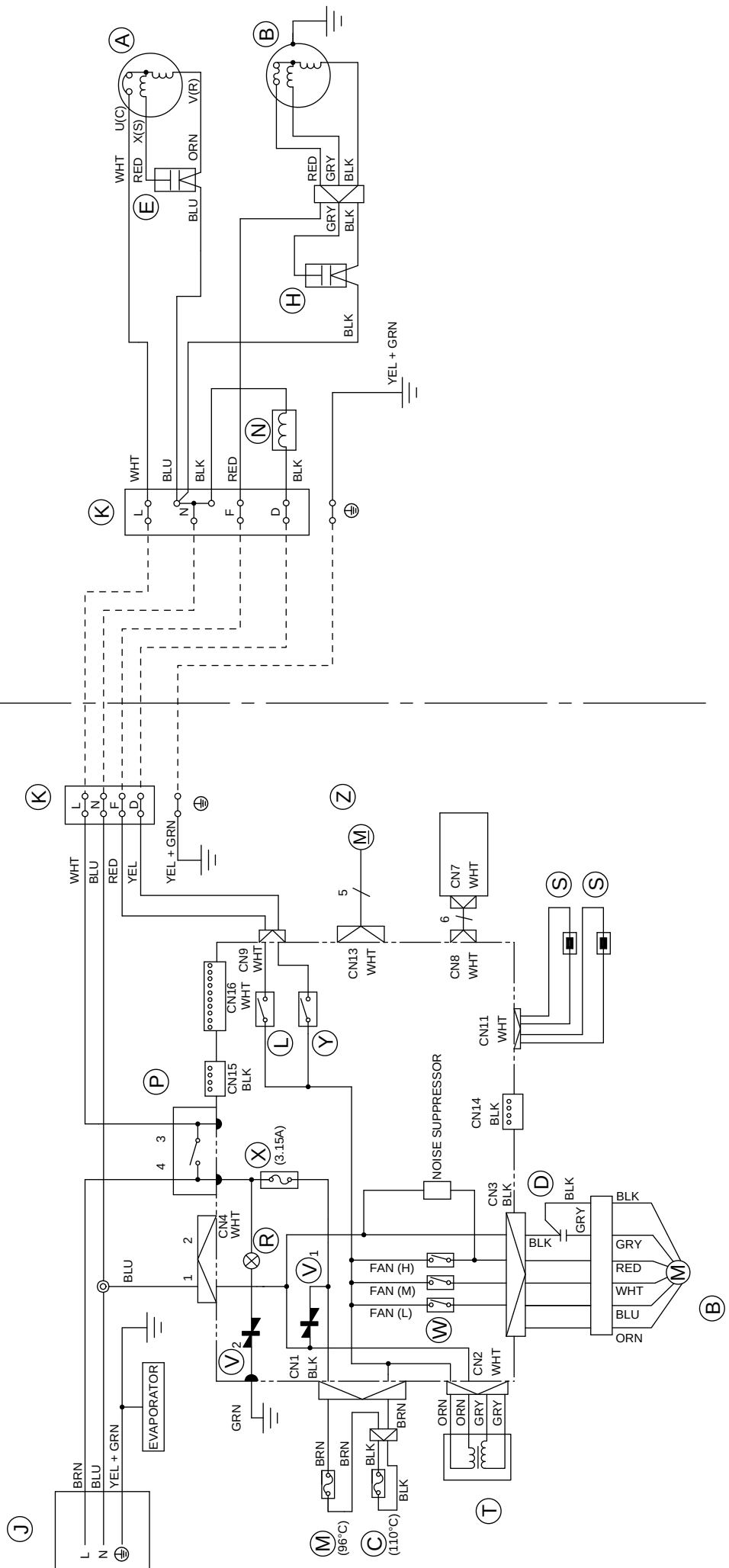
BLK : BLACK



**MODEL RAS-07BH1 / RAC-07BH1
RAS-09BH1 / RAC-09BH1**

INDOOR UNIT

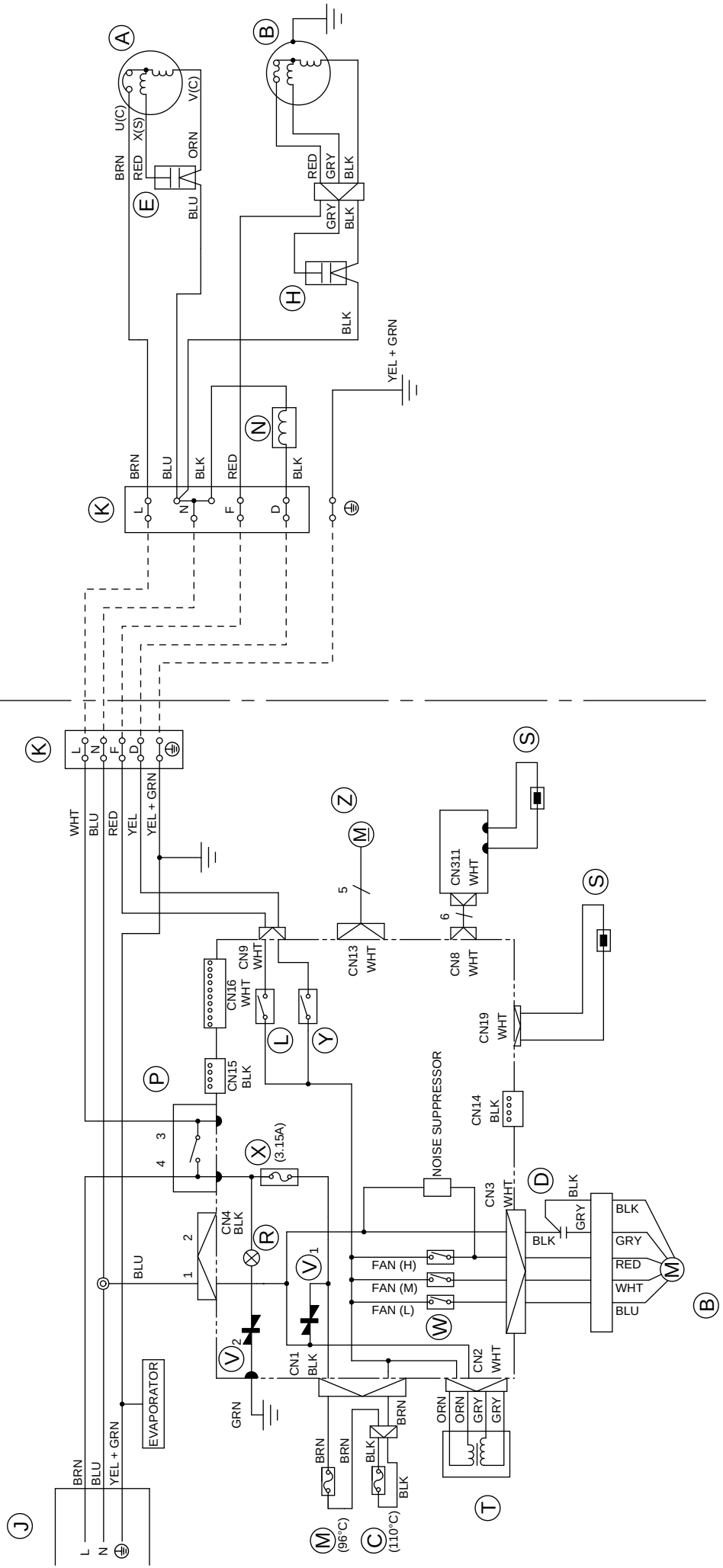
OUTDOOR UNIT



MODEL RAS-14BH1 / RAC-14BH1

INDOOR UNIT

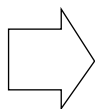
OUTDOOR UNIT



Common, etc.

Q15

There is a difference between the room temperature setting and actual room temperature.

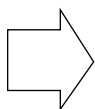


A15

There may be a difference between the room temperature setting and actual room temperature on account of the room structure, air flow, etc.
If there is a difference from the room temperature, adjust the set temperature to keep living space at a comfortable temperature.

Q16

What will happen if the time setting is changed while in a timer operation?



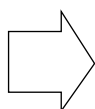
A16

A timer operation is performed until the time after changing the time setting.

Wireless remote controller

Q17

- 1) When the "Automatic" operation mode is selected, "Automatic" does not change by pressing the fan speed select button.
- 2) The room temperature setting is not displayed.
- 3) Pressing the room temperature control button develops transmit mark "👁" and sounds a receive sound but does not display the room temperature setting.



A17

- 1) When the operation mode is "Automatic", the fan speed is automatically fixed to "Automatic".
- 2) At an "Automatic" operation, the room temperature setting is not displayed.
The room temperature is automatically set as follows.

At cooling

Set at 27°C.

At dehumidifying

Set to a temperature slightly lower than the room temperature.

At heating

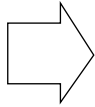
Set at 23°C.
- 3) At an "Automatic" operation, the room temperature setting is not displayed.
However, every pressing "Λ" or "V" button changes 1°C within the range of:
27±3°C when cooling
23±3°C when heating

The room temperature sensing thermistor in the indoor unit detects the room temperature and, according to the particular temperature, automatically performs "Cooling" or "Dehumidifying" operation.
The value indicated not at an "Automatic" but manual operation is not the actual room temperature but the room temperature setting.

Wireless remote controller

Q18

When the room temperature setting is "16", pressing the room temperature control button " V " causes no transmission. At "32", pressing " Λ " causes no transmission either.

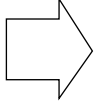


A18

The room temperature is settable within the range of 16 - 32 and not beyond.

Q19

In spite of timer "Preprogram", the time setting is extinguished.



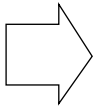
A19

Isn't the time over the preprogrammed time?
As soon as the preprogrammed time is reached, the time setting disappears.

When setting the current time, its indication blinks for approximately 3 minutes.

Q20

After selecting a "Dehumidifying" operation mode, the fan speed mode remains "LO" fan speed".



A20

At a "Dehumidifying" operation, the "LO" fan speed" is forcibly selected.

Timer-Lamp, break-down checking in blinking sign.

Check the break-down factor from the frequency of timer-lamp blinking.

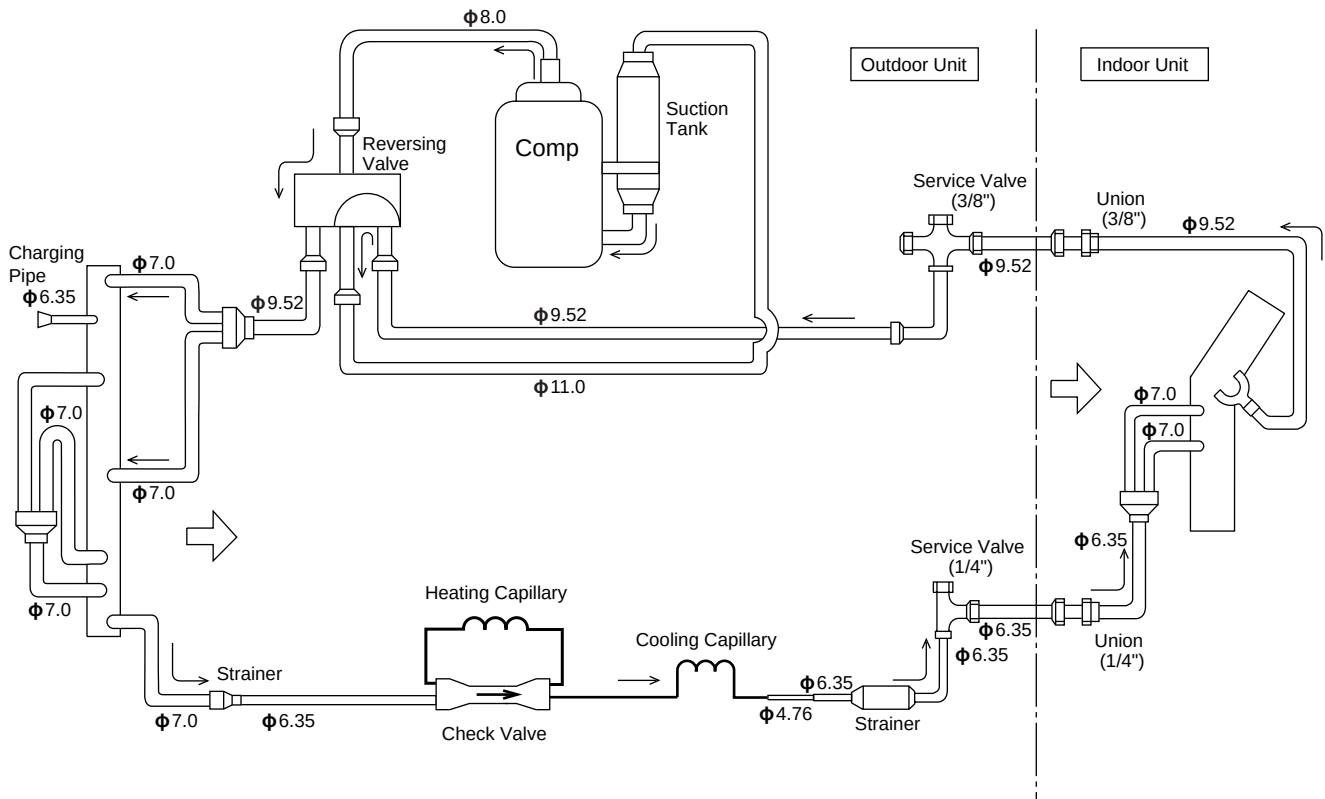
No.	Mode of Timer-Lamp blinking	Indication Factor	Estimated Break-Down Part
1	 1 time	4-way valve not working Inside temperature is low in heating operation time or inside temperature is high in cooling operation time.	(1) 4-way valve is not working. (2) Heat-exchanger thermistor is in disconnection. (Only heating time)
2	 2 times	Outdoor unit operating by compulsion Outdoor unit is under forcible operation or under balancing after forcible operation.	Outdoor electrical goods.
3	 13 times	IC 401 Data read wrongly In case that data read from IC401 is wrong.	IC401 is not in order.
4	 14 times	Heat exchanger thermistor error Heat exchanger thermistor open or short-circuit detected.	(1) Thermistor (2) Indoor - control circuit board.
5	 15 times	Room thermistor error Room thermistor error open or short-circuit detected.	(1) Thermistor (2) Indoor - control circuit board.

( -- 0.5 second on, 0.5 second off.)

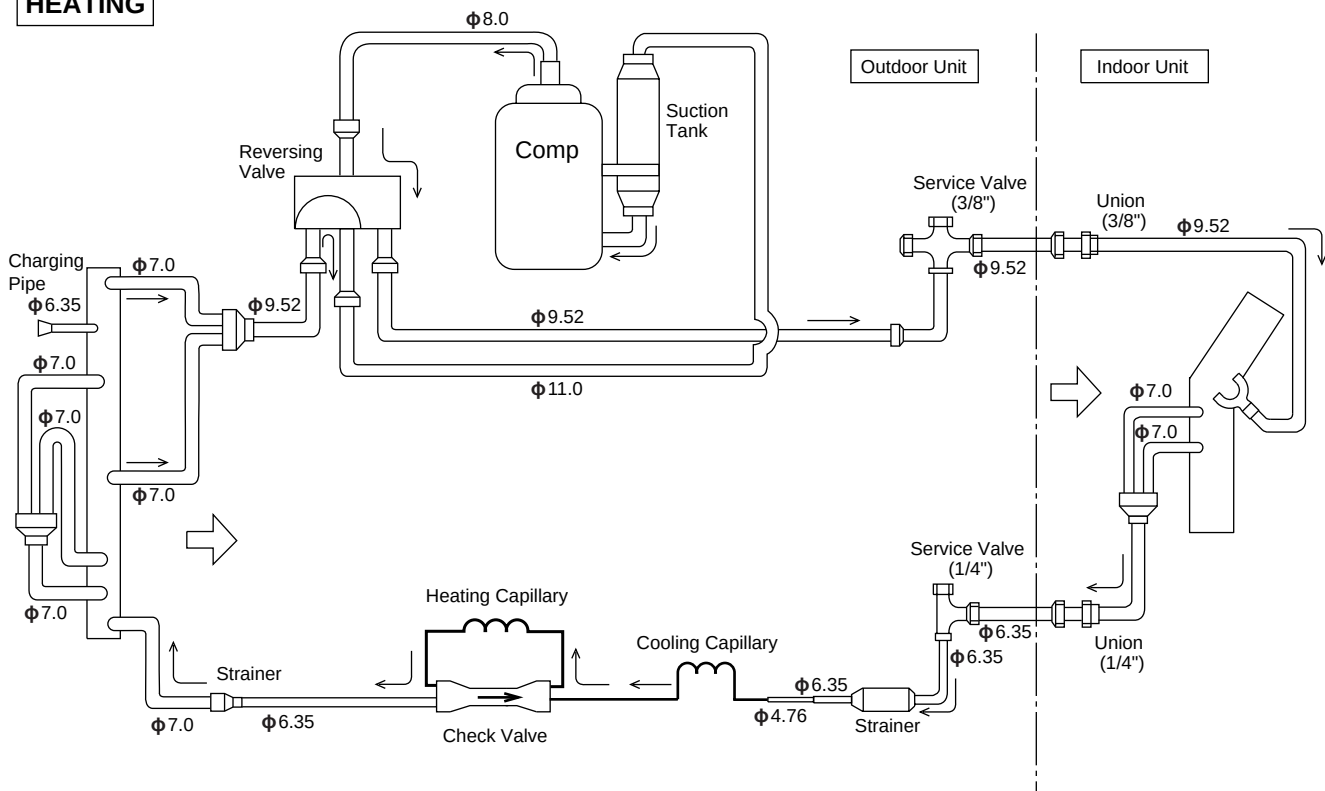
REFRIGERATION CYCLE

MODEL RAS-07BH1 / RAC-07BH1
RAS-09BH1 / RAC-09BH1

COOLING • DEHUMIDIFYING • DEFROST



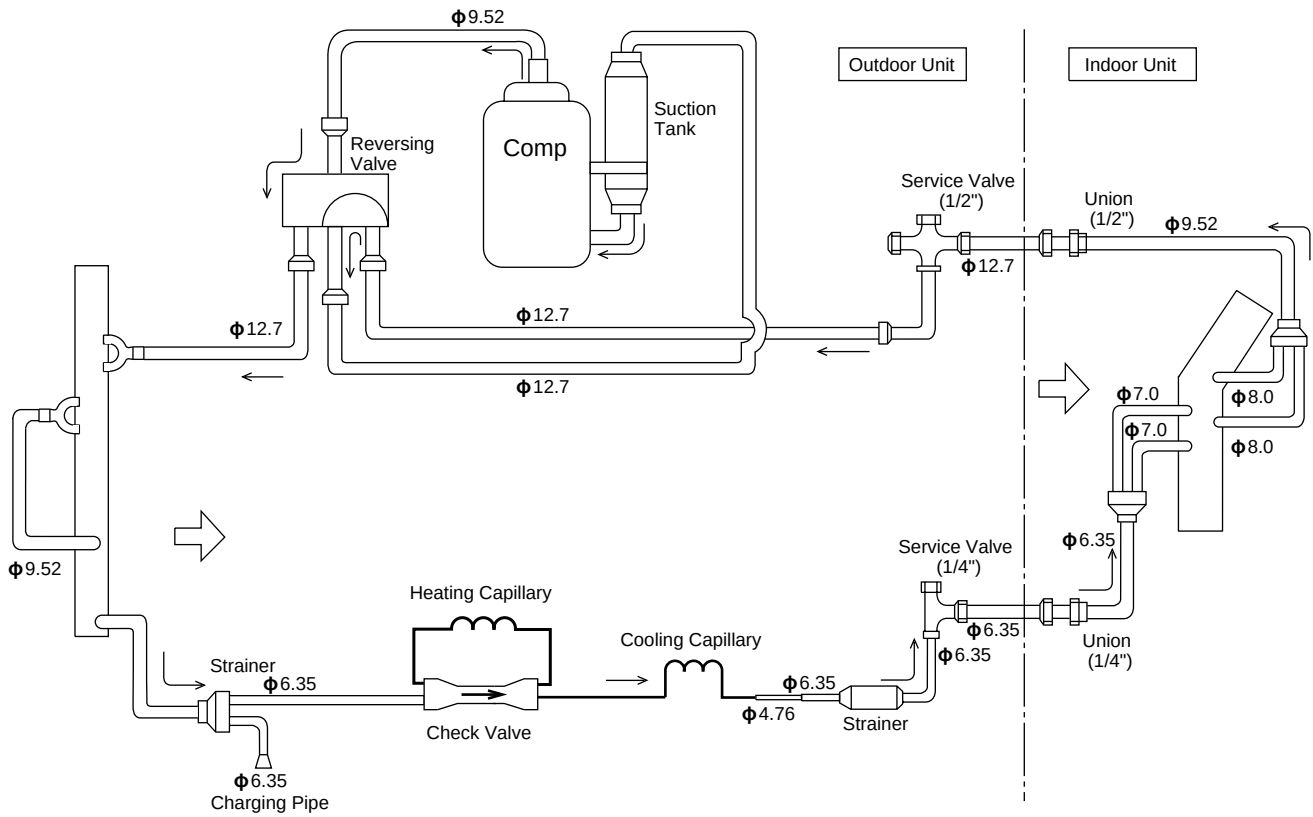
HEATING



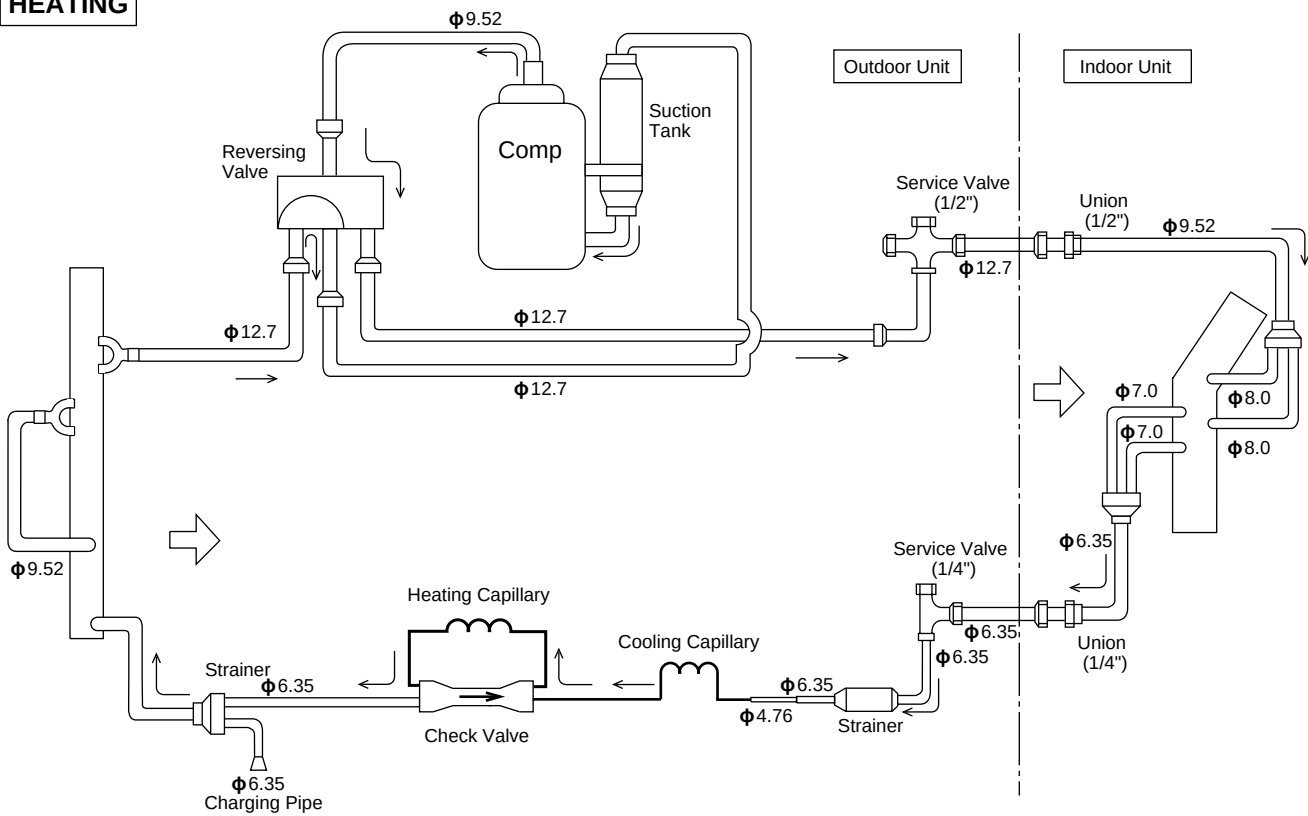
REFRIGERATION CYCLE

MODEL RAS-14BH1 / RAC-14BH1

COOLING • DEHUMIDIFYING • DEFROST

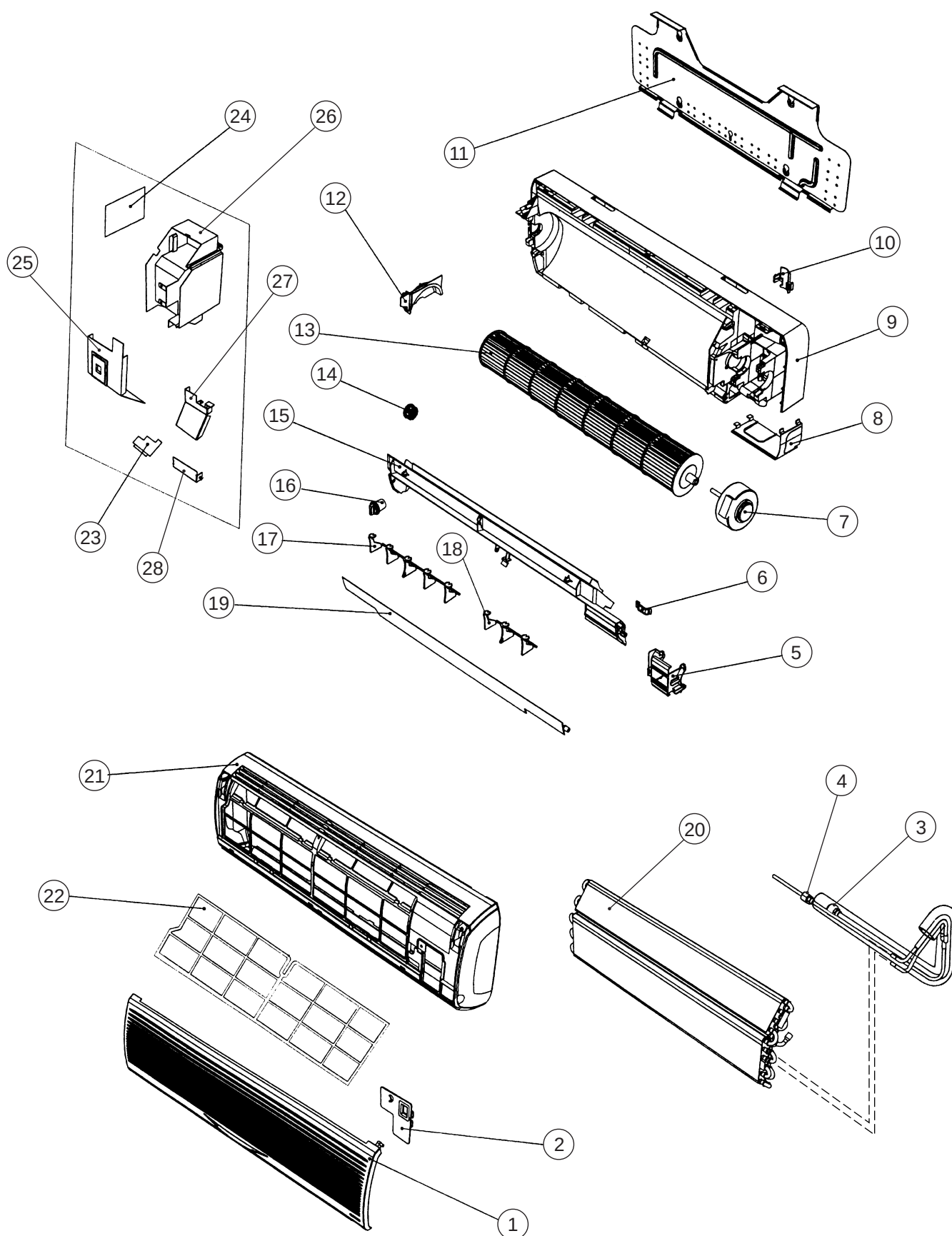


HEATING



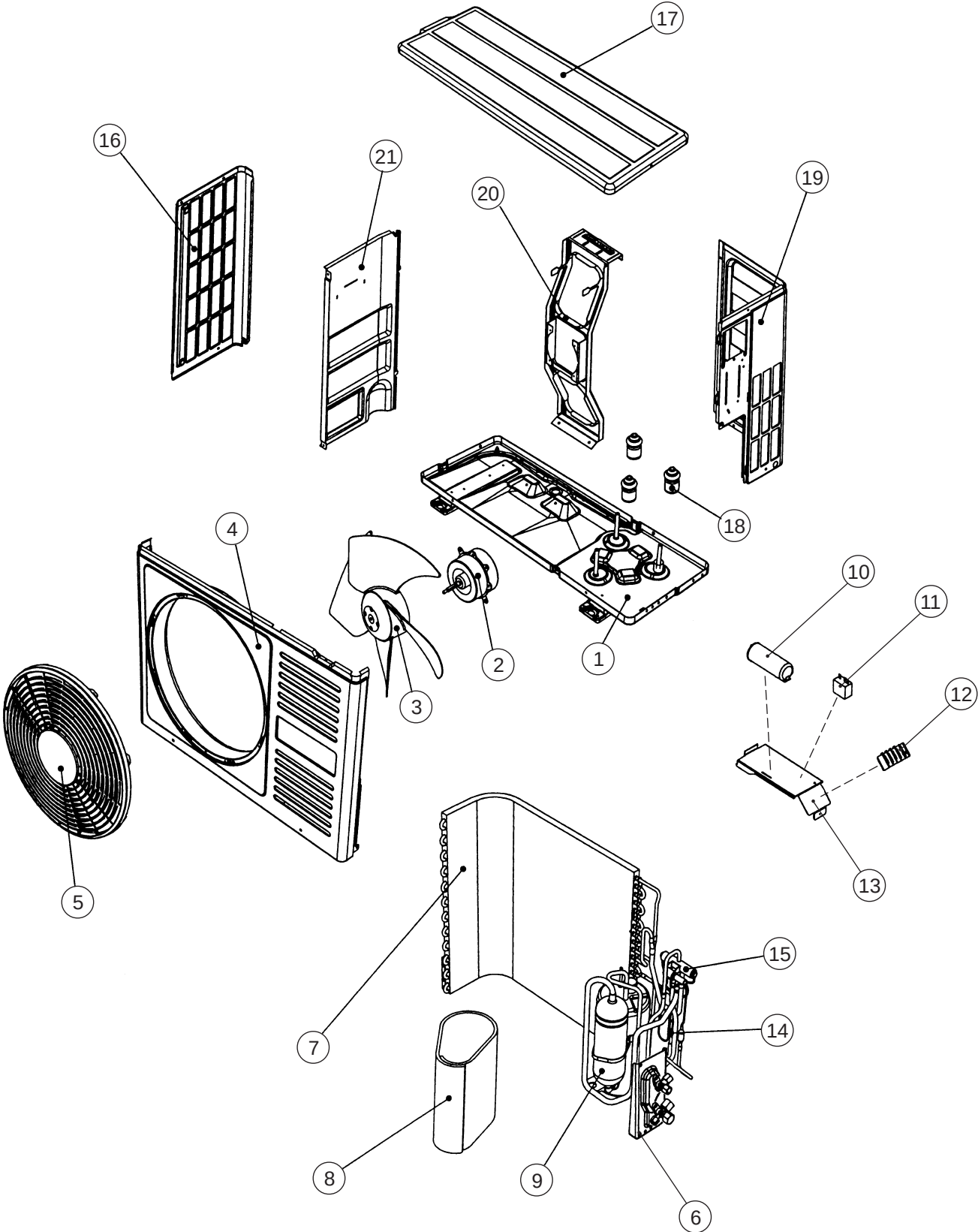
PARTS LIST AND DIAGRAM

MODEL RAS-07BH1, RAS-09BH1



MODEL RAS-07BH1, RAS-09BH1

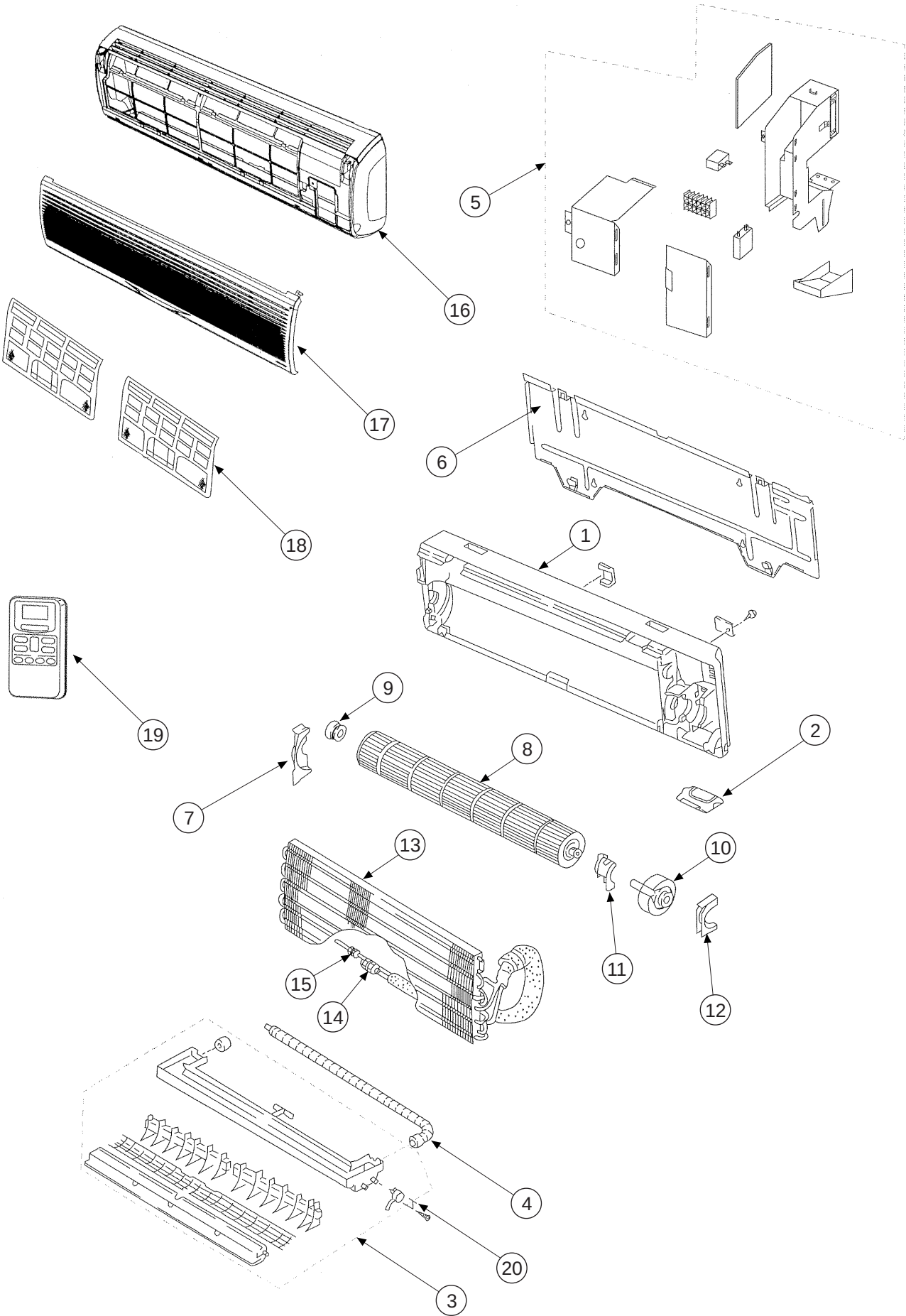
NO.	PART NO. RAS-07BH1, RAS-09BH1	Q'TY / UNIT	PARTS NAME
1	HWRAS-07BH1 901	1	FRONT PANEL ASS'Y
2	HWRAS-07BH1 902	1	ELECTRIC PARTS COVER2
3	HWRAS-07BH1 903	1	3/8FNAHO
4	HWRAS-07BH1 904	1	1/4FNAHO
5	HWRAS-07BH1 905	1	FAN MOTOR SUPPORT
6	HWRAS-07BH1 906	1	BAND
7	HWRAS-07BH1 907	1	FAN MOTOR
8	HWRAS-07BH1 908	1	S-COVER-R
9	HWRAS-07BH1 909	1	CABINET
10	HWRAS-07BH1 910	1	UPPER COVER
11	HWRAS-07BH1 911	1	MOUNTING PLATE
12	HWRAS-07BH1 912	1	FAN COVER
13	HWRAS-07BH1 913	1	TANGENTIAL AIR FLOW FAN
14	HWRAS-07BH1 914	1	FAN SUPPORT ASS'Y
15	HWRAS-07BH1 915	1	DRAIN PAN
16	HWRAS-07BH1 916	1	DRAIN CAP
17	HWRAS-07BH1 917	1	VERTICAL AIR DEFLECTOR-L
18	HWRAS-07BH1 918	1	VERTICAL AIR DEFLECTOR-R
19	HWRAS-07BH1 919	1	HORIZONTAL AIR DEFLECTOR
20	HWRAS-07BH1 920	1	EVAPORATOR ASS'Y
21	HWRAS-07BH1 921	1	FRONT COVER
22	HWRAS-07BH1 922	1	AIR FILTER
23	HWRAS-07BH1 923	1	P.W.B.(INDICATION)
24	HWRAS-07BH1 924	1	P.W.B.(MAIN)
25	HWRAS-07BH1 925	1	ELECTRIC PARTS PLATE
26	HWRAS-07BH1 926	1	ELECTRIC PARTS COVER
27	HWRAS-07BH1 927	1	ELECTRIC PARTS COVER (FRONT)
28	HWRAS-07BH1 928	1	COVER (TERMINAL)



MODEL RAC-07BH1, RAC-09BH1

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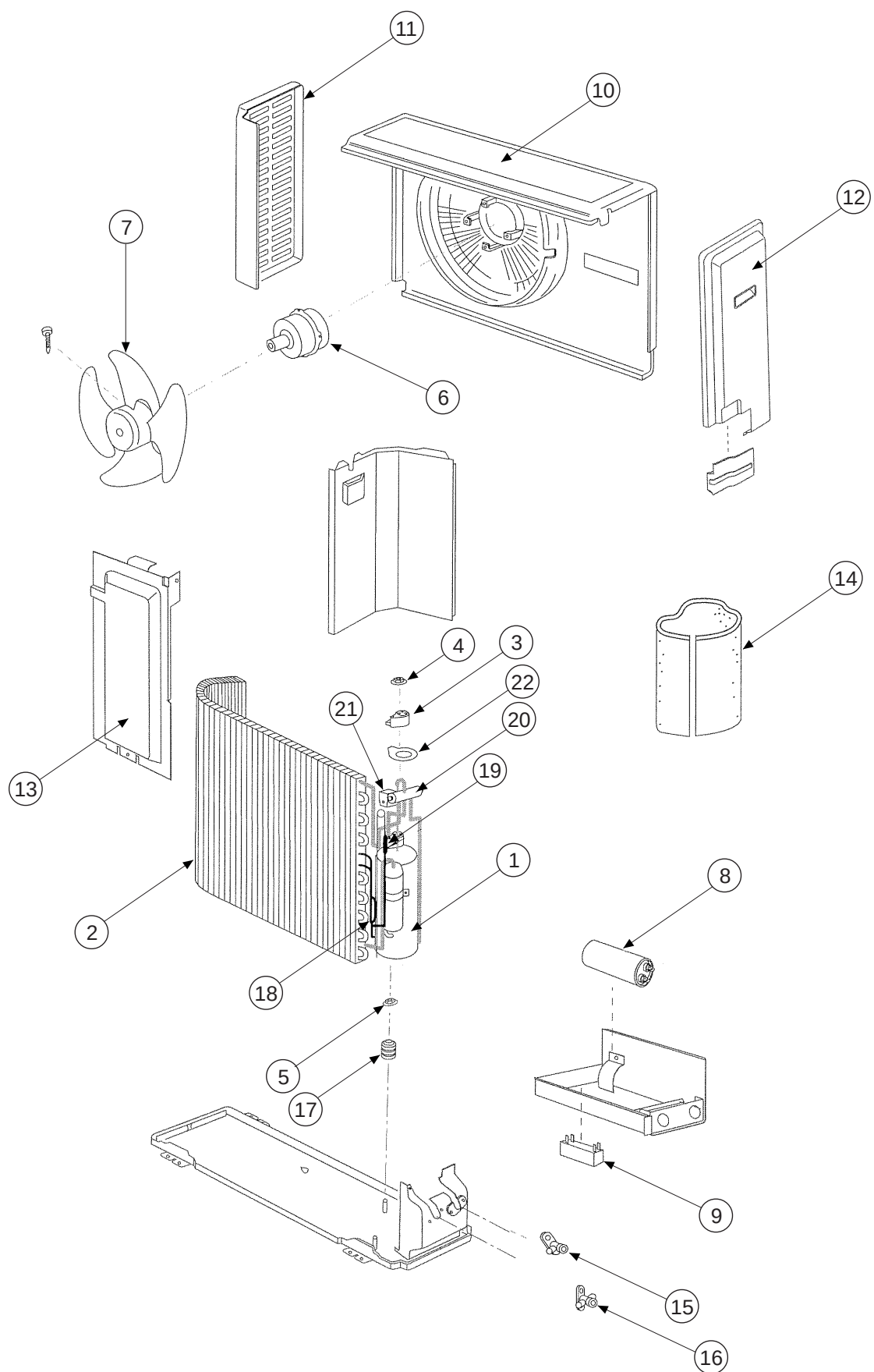
MODEL RAS-14BH1



MODEL RAS-14BH1

NO.	PART NO. RAS-14BH1		Q'TY / UNIT	PARTS NAME
1	RAS-14BH1	001	1	CABINET
2	RAS-14BH1	002	1	COVER(LOWER)(R)
3	RAS-14BH1	003	1	DRAIN PAN
4	RAS-14BH1	004	1	DRAIN HOSE
5	RAS-14BH1	005	1	ELECTRIC BOX ASS'Y
6	RAS-14BH1	006	1	MOUNTING PLATE
7	RAS-14BH1	007	1	FAN COVER
8	RAS-14BH1	008	1	TANGENTIAL AIR FLOW FAN
9	RAS-14BH1	009	1	FAN SUPPORT ASS'Y
10	RAS-14BH1	010	1	FAN MOTOR
11	RAS-14BH1	011	1	FAN MOTOR COVER(L)
12	RAS-14BH1	012	1	FAN MOTOR COVER(R)
13	RAS-14BH1	013	1	EVAPORATOR ASS'Y
14	RAS-14BH1	014	1	1/2" UNION
15	RAS-14BH1	015	1	1/4" UNION
16	RAS-14BH1	016	1	FRONT COVER ASS'Y
17	RAS-14BH1	017	1	GRILL
18	RAS-14BH1	018	2	AIR FILTER
19	RAS-14BH1	019	1	REMOTE CONTROL UNIT
20	RAS-14BH1	020	1	AUTO SWEEP MOTOR

MODEL RAC-14BH1



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This image shows a full page of a document template designed for handwritten notes or essays. It features approximately 30 evenly spaced, thin horizontal grey lines across the entire width of the page. The margins are consistent on all sides, providing ample space for writing. There are no pre-printed questions, headings, or other markings on the page.

HITACHI

RAS-07BH1 / RAC-07BH1
RAS-09BH1 / RAC-09BH1
RAS-14BH1 / RAC-14BH1

AW NO. 0016E